A Complexity-Theoretic Foundation for Metrology

Book II: The Attractor-Basin Program for Type B Certification

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Abstract

This second volume extends the complexity-theoretic framework of Book I into the domain of **Type B measurements**—where knowledge arises from models, constraints, and expert judgement rather than statistical sampling. We present the *Attractor Basin Equivalence Theorem (ABET)* as the exact structural criterion for deterministic certifiability of tail properties, unifying the epistemic notion of a short proof with the physical reality of a reachable, invariant basin. Along the way, we establish the *Minimality Theorem* for basin invariance, the per-step *Double-Bound* method for certifying belief-state updates, and the *Minimal Certification Standard* that forces the unique Observation–Adaptation–Restitution anatomy of any certifiable process. This book offers a rigorous bridge between the formal world of complexity theory and the practical challenges of designing measurement systems that meet the strongest possible certification standards.

Foreword and Guide to This Collection

The first volume built a new foundation for **Type A** measurement—statistical sampling—by showing that deterministic certification demands a finite-state, information-losing structure bounded by guard-bands. This second book continues the story in a different landscape: **Type B** measurements, where the raw randomness of repeated sampling is replaced by the persistent structure of a model.

Here, the key question is: *What physical and algorithmic structure makes a Type B claim certifiable in the same strong sense as in Book I?* The answer leads us to *basins*—regions of belief-state space that, once entered, guarantee future safety—and to a tight correspondence between such basins and certifiable tail predicates.

As in the first volume, the path to this result is neither linear nor purely theoretical. We begin with a concrete case study, abstract its key components into a formal model, and only then push the theory to its necessary and sufficient conditions. The narrative alternates between **constructive examples** and **formal equivalences**, ensuring that each definition earns its keep in proving a major theorem. On its way, developing theory gives rise to the most widely accepted notions and objects in metrology.

The reader is encouraged to follow the chapters in the order presented. Part I develops per-step certification and its complexity bounds. Part II establishes the structural machinery of basins and proves the central ABET equivalence. Part III connects Type B results back to Type A and analyzes the anatomy of a certifiable measurement in its simplest canonical form.

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Part I
From Examples to Per-Step Certification

This first part lays the groundwork for the Attractor-Basin Program. We start with the motivating problem: how can we hope to deterministically certify the long-term behaviour of a system driven by structured, non-statistical information? The journey begins with a realistic Type B case study and proceeds to a formal belief-state model. We then prove that per-step testability is equivalent to controlled information loss, and develop the *Double-Bound* method that will anchor the rest of the book.

Chapter 1: The Quest for Deterministic Certification in Metrology

Introduces the challenge of $\delta \to 0$ certainty for Type B claims, the Π^0_2 barrier, and the roadmap: per-step certification first, tail certification via structural constraints.

Chapter 2: The Outlier-Robust Kalman Filter—A Primer on Type B Certification

A worked example of a belief-state system with robust, polynomial-time updates that lose information each step. Serves as a touchstone for the general theory.

Chapter 3: The Formalism of Belief-State Metrology

Defines the belief-state Θ_t, its compact state space, and the deterministic *Emit/Update* cycle. Introduces multiple equivalent notions of per-step information loss.

Chapter 4: Per-Step Testability and its Equivalence Proof

Shows that per-step deterministic certification is equivalent to controlled information loss. Presents the *Double-Bound* method: a numerical contraction bound plus a refinement procedure ensuring complexity feasibility. Introduces *epistemic losslessness* for basin interiors.

Part II
Basins and Tail Certification

With per-step certification in hand, we turn to the structural problem: how to make claims about the infinite tail of a measurement sequence. The central objects here are *basins*—regions of the belief-state space that guarantee safety once entered. We establish the minimal geometric condition that makes basins invariant, and then prove the *Attractor Basin Equivalence Theorem*, which characterises exactly when a tail predicate is certifiable.

Chapter 5: The Minimal Condition for Invariant Basins—Local Contraction

Axiomatizes basins and proves the *Minimality Theorem*: uniform strict local contraction is both necessary and sufficient for invariance under mild conditions, making it the minimal geometric regularity for tail safety.

Chapter 6: The Attractor Basin Equivalence Theorem (ABET)

The main result: a tail predicate is deterministically certifiable *iff* there exists a reachable, invariant basin meeting emission safety and verifiability conditions. Establishes the exact bridge between logical certifiability and physical structure.

Part III
Unification and Anatomy

The final part unites the Type A and Type B worlds and examines the internal anatomy of a certifiable measurement. We show how Type B dynamics can be cast as guard-band predicates recognisable in the Type A framework, and how the *Minimal Certification Standard* forces a unique decomposition of any certifiable process into its three canonical aspects. The *Simple Test* emerges as the maximally compressed, canonical form of certifiable behaviour.

Chapter 7: Unification of Type A and Type B Measurements

Introduces the Long-Trajectory Cast to express Type B dynamics as guard-band predicates, unifying both measurement types in the same logical framework.

Chapter 8: Analyzing the Testable Measurement

Defines the *Minimal Certification Standard* $\{\varepsilon_n\}$, which induces equivalent state, input, and output guard-bands. Proves the unique decomposition into **Observation**, **Adaptation**, and **Restitution**. Shows that, in the *Simple Test*, all post-entry certified outputs collapse to a single epistemic tick.

Welcome to the Attractor-Basin Program—the second step in a unified theory of certifiable measurement.

Chapter 1: The Quest for Deterministic Certification in Metrology

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Metrology is the science of confidence. It provides the tools to quantify our uncertainty and to make statements of knowledge with a specified, high degree of belief. But can we ever move beyond confidence to certainty? This chapter introduces the motivation for a theory of absolute, zero-error proof for measurement systems, particularly those driven by non-statistical, or Type B, information. We will explore the formidable logical barriers to such proofs and outline the roadmap our theory will follow to overcome them.

1 From Confidence to Proof: The Aspiration for $\delta = 0$

The GUM (Guide to the expression of uncertainty in measurement) provides a powerful and internationally accepted framework for quantifying measurement uncertainty. At its heart lies a probabilistic worldview. A typical result from a Type A analysis, derived from statistical methods, might be a confidence interval for a measurand, accompanied by a statement of confidence, $1 - \delta$. For example, we might state that "the true value lies in the interval $[1.001, 1.003]$ with 95% confidence," which corresponds to a probability of error $\delta = 0.05$.

This is the language of **statistical confidence**. It is powerful, practical, and honest about the limitations of knowledge derived from finite data.

However, a fundamental theoretical question remains: can we ever achieve **logical certainty**? Is it possible to make a statement about a measurement system that is not merely probable, but provably, absolutely true? Can we, in principle, construct a zero-error guarantee, a state of knowledge where $\delta = 0$?

This is the quest that motivates this textbook. We seek to understand the conditions under which it is possible to move from a statement of confidence to a statement of proof. As we will see, this seemingly small leap from a tiny $\delta > 0$ to an absolute $\delta = 0$ is not a simple quantitative step, but a massive qualitative jump that requires a completely different theoretical apparatus. The primary obstacle is the challenge of making claims about the infinite future.

1.1 The Π^0_2 Barrier: Why "Forever" is Hard

Consider a claim about the long-term stability of an instrument, such as:

"For all future times t, the instrument's bias will remain within the bounds $[-\epsilon, \epsilon]$."

This statement contains a universal quantifier ("for all t") over an infinite set (the future). In computability theory and logic, such statements belong to high levels of the "arithmetical hierarchy." A statement of the form $\forall x \exists y, R(x, y)$, where R is a decidable relation, is known as a Π_2^0 predicate.

The crucial fact for our work is that problems defined by such predicates are provably hard. They are not in **NP**, meaning there is no general way to provide a "short," universally checkable proof for them. A finite witness cannot, in general, certify a claim about an infinite number of future events. This is the logical barrier we face.

2 The Failure Modes of Naive Certification

If we attempt to build a deterministic, zero-error certification scheme for a measurement process, we immediately run into two fundamental failure modes that are instances of this Π_2^0 barrier.

2.1 Failure Mode 1: Hidden Randomness

Consider a model of a dynamic measurand, such as a Kalman filter tracking a moving object, where the system's state Θ_t is updated with some internal, unobserved process noise ρ_t:

$$\Theta_{t+1} = U(\Theta_t, \rho_{t+1})$$

Here, ρ_t represents fresh randomness injected into the system at each step that is *not* part of the observable measurement. To certify a claim about the infinite future of such a system, a prover would need to provide a witness that constrains the entire infinite sequence of hidden random variables $(\rho_1, \rho_2, \dots)$. No finite witness can do this. The uncertainty is ongoing and unresolvable, and the certification problem remains intractably hard.

2.2 Failure Mode 2: Lossless Updates

Now consider a simpler model for a static measurand, where all new randomness arrives via an observable measurement z_{t+1}. The belief state is updated via a deterministic function:

$$\Theta_{t+1} = U(\Theta_t, z_{t+1}^{\text{in}})$$

This seems more promising. However, if the update map U is **lossless** (i.e., injective), it preserves all the information from its inputs. The complexity of the belief-state Θ_t grows with each step, as it accumulates the information from the entire history of measurements $(z_1, \dots, z_t)$. After n steps, the state Θ_n is a complex object representing n measurements.

Again, no finite-bit witness can fully describe this ever-growing state. Any attempt to certify a tail property will fail because the verifier can never be sure that some un-witnessed detail from the distant past won't affect the distant future. This, too, results in a co-**NP** or Π_2^0-hard problem.

3 A Roadmap to a New Theory

The failure of these naive approaches points the way forward. To build a theory of deterministic certification, we must impose structural constraints on our measurement model that explicitly break the Π_2^0 barrier. Our theory will be built upon three foundational principles.

1. **Exposing All Randomness.** We will restrict our analysis to systems where all new randomness at each time step enters *only* through the publicly observable input measurement z_t^{in}. There can be no hidden process noise.

2. **Forcing Per-Step Information Loss.** We will require that the deterministic update map $U(\Theta_t, z_{t+1}^{\text{in}})$ is **information-losing**. It must be a non-injective, contractive map that compresses the information from the prior belief Θ_t and the new input measurement z_{t+1}^{in} into a new belief-state Θ_{t+1} that "forgets" some details. This ensures the belief-state remains finite and manageable.

3. **Identifying the Absorbing Belief-Set.** We will show that even with the first two conditions, proving an infinite-tail property requires one final ingredient: identifying an **attractor basin**. This is a "safe" region of the belief-space which, once entered, can never be left. A witness that certifies entry into this basin provides the finite proof for the infinite claim.

4 Outline of the Book

This textbook will systematically develop this theory.

- **Part I: Tutorial and Motivation.** This chapter and the next, a primer on the Outlier-Robust Kalman Filter, will motivate the theory and introduce its core concepts intuitively.

- **Part II: Foundations of Type B Certification.** We will build the formal machinery of belief-states, emit/update maps, and information loss, proving that per-step certification is possible if and only if the update map is contractive.

- **Part III: Attractor Basins and Tail Certification.** We will introduce the attractor basin and prove our main result: the Attractor-Basin Equivalence, which provides the necessary and sufficient conditions for infinite-tail certification.

By the end, we will have constructed a complete and rigorous answer to our initial question, providing a new foundation for understanding and validating Type B measurements.

Chapter 2: The Outlier-Robust Kalman Filter: A Primer on Type B Certification

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Before diving into abstract theory, it is instructive to explore a concrete, sophisticated measurement algorithm that embodies the challenges and concepts of Type B certification. The Outlier-Robust Kalman Filter (ORKF) is a perfect case study. It is a system driven by non-statistical prior knowledge, it incorporates random measurements via a deterministic update rule, and its analysis reveals a deep distinction between per-step validation and infinite-tail proof. This chapter will use the ORKF to build our intuition and demonstrate the need for the formal theory to come.

1 The Setting: Kalman Filtering with Unknown Outliers

The standard Kalman filter provides an optimal state estimate for linear systems under the assumption of Gaussian noise. We consider the more realistic scenario where this assumption is violated due to outliers.

1.1 The State-Space Model

A linear dynamical system is described by:

$$x_t = Ax_{t-1} + w_t \tag{1}$$

$$z_t = Cx_t + v_t \tag{2}$$

where $x_t \in \mathbb{R}^d$ is the hidden state and $z_t \in \mathbb{R}^m$ is the observable measurement. The matrices A and C are known. The process noise $w_t \sim \mathcal{N}(0, Q_w)$ is Gaussian with known covariance Q_w.

1.2 The Type B Problem: Non-Gaussian Measurement Noise

The standard filter assumes $v_t \sim \mathcal{N}(0, R_v)$ with a known, constant covariance matrix R_v. We relax this. We assume the measurement z_t is contaminated by outliers, which can be modeled by assuming the noise v_t is drawn from a Student's-t distribution. A Student's-t

distribution can be represented as a Gaussian scale mixture, where the covariance matrix itself is a random variable.

This means at each time step t, the noise v_t follows a Gaussian distribution with an *unknown* precision matrix (inverse covariance) S_t:

$$v_t \sim \mathcal{N}(0, S_t^{-1})$$

Our knowledge about S_t is not derived from repeated observations (Type A), but from a prior belief (Type B). This is the core of the problem.

2 The ORKF Update: A Type B Belief System

The Outlier-Robust Kalman Filter, as detailed by Agamennoni et al. (2011), addresses this by placing a prior on the unknown precision matrix S_t and estimating it along with the state x_t.

2.1 The Wishart Prior

A conjugate prior for the precision matrix of a multivariate normal distribution is the Wishart distribution. We model our Type B knowledge by assuming that at each step t, our prior belief about S_t is:

$$S_t \sim \mathcal{W}(\nu, V)$$

where ν are the degrees of freedom and V is the scale matrix. These hyperparameters, (ν, V), encode our non-statistical belief about the likely structure of the measurement noise. For simplicity in the following analysis, we will follow a common variant where the prior is placed on the expected value of S_t^{-1}, which is related to the Wishart parameters.

2.2 The Iterative Update Loop

The presence of the unknown S_t makes a direct analytical solution for the posterior intractable. The ORKF instead computes a Maximum A Posteriori (MAP) estimate by iteratively solving a fixed-point equation.

At each time t, given the predicted state $(\hat{x}_{t|t-1}, P_{t|t-1})$ and the new measurement z_t, the algorithm finds the optimal noise precision matrix, let's call it $\Gamma_t^* = S_t$, by iterating the following map F to a fixed point $\Gamma_t^* = F(\Gamma_t^*)$:

$$\Gamma_t^{(i+1)} = F(\Gamma_t^{(i)}) := \frac{sR + \delta(\Gamma_t^{(i)})\delta(\Gamma_t^{(i)})^T + C^T\Sigma(\Gamma_t^{(i)})C}{s+1} \tag{3}$$

where s and R are parameters derived from the Wishart prior, and $\delta(\cdot)$ and $\Sigma(\cdot)$ are complex functions involving the Kalman gain and updated state covariances, which themselves depend on the current estimate $\Gamma_t^{(i)}$.

The crucial observation is that once the hyperparameters (s, R) are fixed, this iterative process is a **fully deterministic algorithm**. The only randomness enters via the observed measurement z_t.

3 Per-Step Certification via Contraction

We can now analyze this procedure from a certification perspective. The **belief-state** at time $t-1$ is $\Theta_{t-1} = (\hat{x}_{t-1|t-1}, P_{t-1|t-1})$. The update map U takes (Θ_{t-1}, z_t) and produces Θ_t. This map involves the iterative solver (3). Can we certify this step?

A step is certifiable if the update map U is polynomial-time computable. This holds if the inner fixed-point iteration is guaranteed to converge in a polynomially bounded number of steps. This guarantee, in turn, is provided if the map F is a **strict contraction**.

Theorem 3.1 (Contraction of the ORKF Update)**.** *The fixed-point map $F(\Gamma)$ in (3) is a strict contraction with respect to the spectral norm, i.e., there exists a $\rho < 1$ such that*

$$\|F(\Gamma_1) - F(\Gamma_2)\| \leq \rho \|\Gamma_1 - \Gamma_2\|$$

if the system matrices (A, C) are bounded and the Wishart prior parameter s is chosen to be sufficiently large.

Proof. The core argument is as follows:

Lemma 3.1 (Lipschitz Bound on $\Sigma(\Gamma)$)**.** *Let $P \succ 0$ be fixed and define*

$$\Sigma(\Gamma) \;=\; \big(C^T P\, C + \Gamma\big)^{-1} C^T P\,.$$

Assume $\Gamma_1, \Gamma_2 \succ 0$ satisfy

$$\lambda_{\min}\big(C^T P\, C + \Gamma_i\big) \;\geq\; \lambda_0 > 0 \quad (i = 1, 2).$$

Then

$$\|\Sigma(\Gamma_1) \;-\; \Sigma(\Gamma_2)\| \;\leq\; \frac{\|C^T P\|\,\|C\|}{\lambda_0^2}\,\|\Gamma_1 - \Gamma_2\|.$$

Main theorem proof. Write

$$\Sigma(\Gamma_1) - \Sigma(\Gamma_2) = \big(C^T P\, C + \Gamma_1\big)^{-1}\big(\Gamma_2 - \Gamma_1\big)\big(C^T P\, C + \Gamma_2\big)^{-1} C^T P.$$

Taking operator norms and using $\|(C^T P\, C + \Gamma_i)^{-1}\| \leq 1/\lambda_0$, we obtain

$$\|\Sigma(\Gamma_1) - \Sigma(\Gamma_2)\| \leq \frac{\|\Gamma_2 - \Gamma_1\|}{\lambda_0^2}\,\|C^T P\|\,\|C\|,$$

as claimed. □

1. The difference $F(\Gamma_1) - F(\Gamma_2)$ can be shown to depend primarily on the difference in the updated state covariances, $\Sigma(\Gamma_1) - \Sigma(\Gamma_2)$, scaled by $\frac{1}{s+1}$.

$$\|F(\Gamma_1) - F(\Gamma_2)\| \leq \frac{\|C\|^2}{s+1}\|\Sigma(\Gamma_1) - \Sigma(\Gamma_2)\| + \text{h.o.t.}$$

2. By Lemma above, the map $\Gamma \mapsto \Sigma(\Gamma)$ is itself Lipschitz continuous. Its Lipschitz constant depends on the norms of the system matrices and the inverse of the matrix $(C^T PC + \Gamma)$.
$$\|\Sigma(\Gamma_1) - \Sigma(\Gamma_2)\| \leq K \cdot \|\Gamma_1 - \Gamma_2\|$$
where K depends on $\|P\|, \|C\|^2$, and a lower bound on the eigenvalues of $(C^T PC + \Gamma)$.

3. Combining these, we get a Lipschitz constant for F of the form:
$$\rho \approx \frac{K\|C\|^2}{s+1}$$

By choosing the Wishart strength parameter s to be large enough relative to the fixed system gains, we can ensure that $\rho < 1$. By the Banach Fixed-Point Theorem, the iteration is guaranteed to converge geometrically to a unique fixed point. The number of iterations needed to reach precision ϵ is $O(\log(1/\epsilon))$, which is polynomial in the number of bits of precision. □

This result is profound. It means that the seemingly complex ORKF update is a **polynomial-time computable, information-losing** map. Therefore, each step is certifiable. A prover can supply the resulting belief-state Θ_t as a witness, and a verifier can re-run the polynomial-time algorithm to check its correctness.

4 The Barrier to Tail Certification

We have established that we can trust each individual step of the filter. But what about a claim concerning the infinite future, such as:

> P_{stable}: "For all future times $k > n$, the trace of the state error covariance, $\mathrm{Tr}(P_{k|k})$, will remain below a threshold β."

This is a claim about the long-term stability of the filter. Can we provide a finite witness for this?

The answer is no, not without more structure. Per-step contractivity is not enough. The sequence of incoming random measurements $z_{n+1}, z_{n+2}, \ldots$ could, in principle, be adversarial and consistently push the state towards the boundary of the "safe" region. Certifying that this will *never* happen requires a guarantee that spans an infinite number of future random inputs. This is the Π^0_2 barrier we identified in Chapter 1.

To bridge this gap, we need a stronger concept: a region in the belief-space which, once entered, can never be escaped, regardless of future measurements. This is the concept of an **attractor basin**, which will be the central subject of Part III. The ORKF example demonstrates perfectly why such a concept is not just a theoretical nicety, but an absolute necessity for proving anything about the long-term behavior of a Type B measurement system.

References

References

[1] G. Agamennoni, J. I. Nieto, and E. M. Nebot. "An outlier-robust Kalman filter." In *2011 IEEE International Conference on Robotics and Automation*, pp. 1551-1558. IEEE, 2011.

Chapter 3: The Formalism of Belief-State Metrology

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The ORKF primer in the previous chapter provided a concrete example of a measurement system driven by non-statistical (Type B) information. To build a general theory, we must now abstract the key components from that example into a universal, formal framework. This chapter introduces the core concepts of our theory: the ***Belief-State*** *as the carrier of all Type B knowledge, and the deterministic* ***Emit*** *and* ***Update*** *maps that govern its evolution. We will then provide a rigorous, multi-faceted definition of the per-step* ***information loss*** *that is the prerequisite for all subsequent certification.*

1 The Belief-State: A Formal Representation of Knowledge

In our framework for Type B certification, we shift perspective from a raw stream of data to the metrologist's evolving state of knowledge. We encapsulate all non-statistical information available at a given time t into a single mathematical object.

Definition 1.1 (Belief-State)**.** A **belief-state**, denoted Θ_t, is an element of a space $\mathcal{M}$ that represents the complete summary of all non-statistical (Type B) information at time t. This includes, but is not limited to:

- Calibration parameters (e.g., offsets, scale factors, polynomial coefficients).
- Manufacturer specifications (e.g., tolerance intervals).
- Constraints derived from physical laws (e.g., conservation principles).
- Parameters of prior distributions from expert judgment (e.g., the hyperparameters of a Wishart prior).

To ensure this object is amenable to finite certification, we impose a crucial structural requirement on the space it inhabits.

Definition 1.2 (Belief-State Space)**.** The belief-state space, $\mathcal{M}$, is a **finite-dimensional, compact** metric space, typically a compact subset of $\mathbb{R}^d$ for some integer d. The space of input measurements, $\mathcal{Z}^{\text{in}}$, and the space of output measurements, $\mathcal{Z}^{\text{out}}$, are also assumed to be compact.

The compactness condition is essential. It ensures that the space is totally bounded, meaning any belief-state Θ_t can be encoded with a finite number of bits to any desired precision. This is a necessary precondition for constructing a finite witness.

2 The Deterministic Two-Step Cycle

Our model assumes that all new randomness enters the system exclusively through an observable measurement z_{t+1}. The evolution of the belief-state is then governed by a deterministic two-step cycle.

Definition 2.1 (Emit and Update Maps)**.** The evolution of the belief-state is defined by two deterministic maps:

1. The **Emit Map**, $E : \mathcal{M} \to \mathcal{Z}^{\text{out}}$, which deterministically produces the system's "ideal" or predicted output $z_{t+1}^{\text{out}} = E(\Theta_t)$ based on the current belief-state Θ_t. We assume E is Lipschitz continuous.

2. The **Update Map**, $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$, which deterministically assimilates the newly observed input measurement z_{t+1}^{in} to produce the next belief-state: $\Theta_{t+1} = U(\Theta_t, z_{t+1}^{\text{in}})$. We assume U is also Lipschitz continuous.

As established in Chapter 1, a key requirement for certification is that the Update map must be **information-losing**. This update map must compress the information contained in the pair $(\Theta_t, z_{t+1}^{\text{in}})$ into a new state Θ_{t+1} that resides in the same compact, finite-dimensional space $\mathcal{M}$.

3 Geometry of Per-Step Information Loss

3.1 Overview of Per-Step Information Loss

Per-step information loss is the phenomenon by which an update map

$$U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$$

fails to preserve all distinguishability or degrees of freedom of its input when producing its output belief state. Different mathematical manifestations of this failure coexist; they are neither identical nor interchangeable without additional structure. In this section we define four distinct notions of per-step information loss, exhibit their intended meanings, and analyze their interrelations. The goal is to (i) make precise what it means for a step to "lose information," (ii) isolate the assumptions under which one form implies another, and (iii) expose the limits and pathologies—especially in the continuous, infinite-precision setting—so that later results can be stated with the minimal necessary hypotheses.

We will work throughout with compact metric spaces $\mathcal{M}$ (belief space) and $\mathcal{Z}^{\text{in}}$ (input space), equipped with a metric d. When needed, finite-resolution quantizations will be introduced explicitly; nothing in this section presumes an a priori discretization except when stated.

3.2 Exact Non-Injectivity

Definition 3.1 (Exact Non-Injectivity)**.** The update map $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ exhibits *exact non-injectivity* if there exist two distinct inputs $(\theta, z^{\text{in}}) \neq (\theta', z'^{\text{in}})$ such that

$$U(\theta, z^{\text{in}}) \;=\; U(\theta', z'^{\text{in}}).$$

Equivalently, U is not one-to-one.

This is the most literal and algebraic form of information loss: different joint input pairs are mapped to the same belief state, causing irrecoverable collapse of distinctions. It underlies classical certificate constructions where a single coarse label (the image) admits multiple preimages, and the ambiguity is the resource enabling succinct witnessing.

Example. Let $\mathcal{M} = [0, 1]$, $\mathcal{Z}^{\text{in}} = \{0, 1\}$, and define

$$U(\theta, z^{\text{in}}) = \begin{cases} \theta/2, & z^{\text{in}} = 0, \\ \theta/2, & z^{\text{in}} = 1. \end{cases}$$

Then $(\theta, 0)$ and $(\theta, 1)$ are distinct but have the same image, so the map is exactly non-injective. The loss is maximal in the sense that the input measurement z^{in} is completely erased by U.

Remark. Exact non-injectivity is a binary property; it does not quantify the "amount" of loss or relate it to scales of indistinguishability. Those refinements appear in the subsequent subsections.

3.3 ε-Collapse: Finite-Resolution Indistinguishability

Definition 3.2 (ε-Collapse)**.** Fix a resolution parameter $\varepsilon > 0$. The update map U exhibits an ε-collapse if there exist two inputs $(\theta, z^{\text{in}}), (\theta', z'^{\text{in}})$ with $d(\theta, \theta') \geq \delta$ for some fixed $\delta > 0$ (i.e., they are distinguishable at scale δ) but whose outputs are within ε:

$$d\big(U(\theta, z^{\text{in}}),\, U(\theta', z'^{\text{in}})\big) \leq \varepsilon.$$

Intuitively, ε-collapse captures the idea that even though two inputs were meaningfully different, the update map crushes their distinction below the resolution threshold ε, rendering them effectively indistinguishable to any observer limited to that precision.

Local versus uniform collapse. One can further distinguish:

- *Existential ε-collapse:* There exists at least one pair of well-separated inputs whose images are ε-close.
- *Uniform ε-collapse:* There is a nontrivial volume of input pairs (θ, z^{in}), $(\theta', z'^{\text{in}})$ with $d(\theta, \theta') \geq \delta$ whose outputs fall into the same ε-ball; this supports metric-entropy lower bounds.

Example. Let $\mathcal{M} = \mathbb{R}$, $U(\theta, z^{\text{in}}) = \theta + z^{\text{in}}$, with $z^{\text{in}} \in [-\varepsilon, \varepsilon]$ drawn from a continuous interval. Then any two inputs (θ, z^{in}) and $(\theta', z'^{\text{in}})$ satisfying $\theta' = \theta + (z^{\text{in}} - z'^{\text{in}})$ produce identical outputs; in particular, distinct θ separated by δ can have output difference arbitrarily small (depending on choices of z), so an ε-collapse occurs for any fixed $\varepsilon > 0$.

Relation to exact non-injectivity. Exact non-injectivity implies ε-collapse for every ε, but the converse need not hold: two distinct inputs may map to outputs within ε without coinciding exactly. Thus ε-collapse is a *quantitative, approximate* weakening of exact non-injectivity.

3.4 Mutual Information Drop at Finite Resolution

We now introduce an information-theoretic measure of per-step loss that operates at finite resolution via quantization.

Definition 3.3 (Finite-Resolution Mutual Information Drop)**.** Let (Θ, Z^{in}) be treated as a joint random variable supported on a finite quantization of $\mathcal{M} \times \mathcal{Z}^{\text{in}}$, induced by a partition into cells of diameter at most ε. Let $\Theta_{t+1} := U(\Theta_t, Z^{\text{in}}_{t+1})$ be the resulting (quantized) output. Define the finite-resolution mutual information drop as the gap

$$\Delta_I^{(\varepsilon)} := H(\Theta_t, Z^{\text{in}}_{t+1}) - I\big((\Theta_t, Z^{\text{in}}_{t+1}); \Theta_{t+1}\big) = H\big((\Theta_t, Z^{\text{in}}_{t+1}) \,\big|\, \Theta_{t+1}\big),$$

where entropies and mutual informations are computed with respect to the chosen finite alphabet arising from the ε-quantization.

The quantity $\Delta_I^{(\varepsilon)}$ is nonnegative, and $\Delta_I^{(\varepsilon)} > 0$ signifies that knowledge of the output Θ_{t+1} does not fully determine the input pair $(\Theta_t, Z^{\text{in}}_{t+1})$: some uncertainty remains, i.e., information is lost.

Interpretation. This form of loss is graded and sensitive to resolution: coarse quantizations may hide fine distinctions (making $\Delta_I^{(\varepsilon)}$ large) while refining ε decreases the artificial loss introduced by quantization. It subsumes ε-collapse in the following loose sense: if two well-separated inputs are mapped to outputs within the same ε-cell (i.e., an ε-collapse), then the fiber over that cell contains multiple quantized preimages, inflating the conditional entropy and hence $\Delta_I^{(\varepsilon)}$.

Example. Suppose Θ_t is uniform over N states and Z^{in}_{t+1} is independent, but U maps all of them to $M < N$ distinct outputs due to many-to-one collapse within the quantization. Then the total entropy in the input exceeds that in the output, and $\Delta_I^{(\varepsilon)} \geq \log_2(N/M) > 0$.

Caveat in continuous limit. If the underlying spaces are continuous and we attempt to let $\varepsilon \to 0$ without care, the discrete entropies may diverge and mutual information can become infinite or ill-defined. The correct procedure is always to fix a finite ε, compute $\Delta_I^{(\varepsilon)}$, and then control its behavior as $\varepsilon \to 0$ under additional assumptions (see Section **??** below).

3.5 Strict Contraction and Its Consequences

Definition 3.4 (Strict Contraction on Belief States)**.** Let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$. We say U is a *strict contraction* on the belief-state component uniformly in the input if there exists $\rho \in [0, 1)$ such that for all $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$ and all $\theta, \theta' \in \mathcal{M}$:

$$d\big(U(\theta, z^{\text{in}}),\, U(\theta', z^{\text{in}})\big) \;\leq\; \rho \cdot d(\theta, \theta').$$

Strict contraction is a geometric form of information collapse: it forces orbits to come closer, shrinking distinctions exponentially over repeated application. However, on a complete metric space, a strict contraction in the classical sense (mapping $\mathcal{M}$ to itself for fixed z) is injective; hence it cannot create exact non-injectivity. This tension is central to what we later call the *epistemic losslessness* phenomenon on basins.

Consequences.

- *Stability of small differences:* If two beliefs start close, their images remain controlled and even closer, with ratio bounded by ρ.
- *Damping of perturbations:* Iterating the contraction on belief alone diminishes any initial discrepancy geometrically, which in a finite-resolution approximation can make distinguishability arbitrarily small.
- *Non-creation of new distinctions:* Contraction cannot separate previously indistinguishable inputs; it only collapses.

Example. Take $\mathcal{M} = [0, 1]$, $\mathcal{Z}^{\text{in}}$ arbitrary, and define $U(\theta, z^{\text{in}}) = \frac{1}{2}\theta + h(z^{\text{in}})$ where h is any bounded function. Then $\rho = 1/2$ works, and for fixed z^{in}, the map in θ is a strict contraction.

Remark on injectivity vs. effective collapse. Although strict contraction does not induce exact non-injectivity, it can—when composed with finite-resolution observation—make formerly distinguishable beliefs indistinguishable at any desired scale. This discrepancy between *global injectivity* and *practical indistinguishability* motivates the later notion of *epistemic losslessness*.

1.7 Pathologies and the Continuous Limit

When the belief and input spaces are continuous (e.g., manifolds or subsets of Euclidean space), naive extensions of discrete-information notions can fail or become ill-behaved. This subsection isolates the key pitfalls and specifies what is required to avoid them.

Infinite mutual information. In the continuous setting, differential entropy and mutual information can be infinite or undefined unless one fixes a quantization or imposes strong regularity. For example, if the joint distribution of $(\Theta_t, Z^{\text{in}}_{t+1})$ concentrates on a lower-dimensional manifold, the Shannon mutual information in any limiting sense may diverge. Therefore, all statements involving information drop must be understood as statements about *finite-resolution approximations* first, with limits taken only under controlled assumptions.

Degeneration without control. Let U be a strict contraction. Absent uniform continuity controls, arbitrarily small differences in input beliefs could be magnified in their effect on finite-resolution statistics after many steps, or conversely, meaningful structural differences could hide under pathological coordinate distortions. Similarly, if the quantization cells are not chosen with respect to the topology induced by d, one might observe spurious ε-collapse artifacts that are purely discretization effects.

Nonuniform behavior. Without uniformity assumptions (e.g., uniform continuity of U in both arguments or compactness of domains), the rate at which $\Delta_I^{(\varepsilon)}$ vanishes as $\varepsilon \to 0$ can vary wildly across the space, precluding any global statement about effective losslessness. This motivates the introduction, in Section **??**, of conditions that ensure uniform control over the quantization-induced loss.

Summary of required safeguards. To avoid these pathologies in subsequent limit arguments we will impose:

- Compactness of $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$,
- Uniform continuity (or Lipschitz bounds) of U in both arguments,
- Explicit finite-resolution quantizations with diameters tied to ε,
- Consistent refinements (nested covers) so that passing to the limit $\varepsilon \to 0$ can be made coherent.

These prerequisites are minimal in the sense that dropping any of them makes the limiting behavior fragile or ill-defined.

3.6 Summary of the Geometry of Per-Step Loss

We defined four distinct formal manifestations of per-step information loss:

- **Exact Non-Injectivity:** algebraic identification of distinct inputs;
- **ε-Collapse:** approximate loss at finite resolution;
- **Mutual Information Drop:** graded, quantitative uncertainty remaining in the input after observing the output, defined via finite-resolution quantization;
- **Strict Contraction:** geometric shrinkage of belief distinctions that, while globally injective on a complete space, can render differences epistemically negligible at finite resolution.

Their interrelations are nuanced: exact non-injectivity implies collapse and information drop, but not vice versa; strict contraction does not inherently introduce exact non-injectivity yet—under uniform control—can drive the observable loss to zero as resolution refines, giving rise to the useful conceptual notion of epistemic losslessness. We also highlighted the necessary technical safeguards in the continuous setting to make limit statements meaningful.

These distinctions and quantified behaviors provide the foundation for the per-step testability results to follow and, in particular, set the stage for the *double-bound* analysis in Chapter 4: one bound numerical (vanishing finite-resolution loss under contraction) and one complexity-theoretic (semi-decidability via enumeration of refinements), which together reconcile the apparent tensions between contraction-based stability and the necessity of information loss for certification.

4 The Path Forward: From Steps to Tails

This chapter has laid the formal groundwork for our theory. We have a **Belief-State Θ_t** that lives in a compact space, an **Update map U** that must be information-losing, and an **Emit map E** that connects belief to observation.

In the next chapter, we will combine these ingredients to prove our first major result: that these conditions are necessary and sufficient to make the certification of each individual step of a Type B measurement a computationally tractable problem. We will then see in the final part of this book how this per-step certifiability is a crucial prerequisite for the much harder problem of certifying properties of the infinite tail.

Chapter 4: Per-Step Testability and its Equivalence Proof

Dmytro Surdu

In Chapter 3, we established the formal machinery of belief-states and defined information loss within a deterministic update step. We now deploy this machinery to prove the first pillar of our theory: the principle of per-step testability. This chapter will demonstrate that the requirement of information loss is both a necessary and sufficient condition for the certification of a single step in a Type B measurement chain to be a computationally tractable problem. We will also formally prove that the five different characterizations of information loss are equivalent for this purpose.

1 The Per-Step Certification Problem

The fundamental building block of trust in a measurement chain is the ability to verify each discrete step. The problem is as follows: given a certified belief-state Θ_{t-1} from the previous step and the new observation z_t, can we verify a prover's claim that the new, correct belief-state is Θ_t?

We formalize this by defining a language for each step. We say per-step certification is possible with *short witnesses* if there is a fixed polynomial p and a polynomial-time algorithm V that can verify the update.

Definition 1.1 (Per-Step Certification with Short Witnesses)**.** A system admits per-step certification if there exists a polynomial p and a verifier V such that for any step t, $V(\Theta_{t-1}, z_t^{\text{in}}, w)$ accepts if and only if w is a correct encoding of $\Theta_t = U(\Theta_{t-1}, z_t^{\text{in}})$, and $|w| \leq p(k)$, where k is the number of bits required to represent the model parameters. The witness size must be independent of the step number t.

2 Per-Step Testability and the Double-Bound Theorem

2.1 Setup and Definitions

We retain the basic objects from Chapter 3. Let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ be the per-step update map on compact metric spaces with metric d. Let $\Theta_t \in \mathcal{M}$ be the belief state at step t and

$z^{\text{in}}_{t+1} \in \mathcal{Z}^{\text{in}}$ the new input, producing

$$\Theta_{t+1} = U(\Theta_t, z^{\text{in}}_{t+1}).$$

We consider the per-step certification problem: given a finite observed head (sequence of past inputs/outputs up to step t) together with a purported certificate about the current belief state, can a deterministic polynomial-time verifier decide that the continuation from this step will satisfy a specified tail property? The fundamental question is what structure on U makes such a one-step certificate possible (i.e., testability), and what obstruction prevents it.

Per-step certificate format. At resolution $\varepsilon > 0$, a natural certificate is a coarse label for the current belief-state together with an index resolving its fiber ambiguity (analogous to the "state witness" in Book I), or—more generally—any succinct description that disambiguates among inputs that would otherwise be indistinguishable at precision ε.

Finite-resolution information loss. Let $\Delta_I^{(\varepsilon)}$ denote the finite-resolution mutual information drop for the single step as defined in Chapter 3. We say the step has *positive information loss at scale* ε if $\Delta_I^{(\varepsilon)} > 0$. This will serve as the quantitative sufficient condition for per-step testability.

Strict contraction region. A region $R \subseteq \mathcal{M}$ is said to admit a *uniform strict contraction* if there is $\rho \in [0, 1)$ such that for all $\theta, \theta' \in R$ and all z^{in}, $d(U(\theta, z^{\text{in}}), U(\theta', z^{\text{in}})) \leq \rho\, d(\theta, \theta')$. Such regions will play a special role in the double-bound analysis below.

Complexity classes. We adopt the standard interpretation: a language L of "good" heads (or state-descriptions) is in **NP** if there exists a polynomial-size certificate verifiable in deterministic polynomial time; we say the certification problem is at most recursively enumerable (r.e.) if there exists a semi-decision procedure that accepts all positive instances (possibly without halting on negatives).

Reference to Book I. The sufficiency and necessity statements for per-step testability were first formalized in Book I under a discrete, loss/non-loss dichotomy. We rephrase and adapt those results here to the present finite-resolution, continuous-ambient setting. Wherever possible, we invoke the original proofs from Book I, with modifications only to accommodate quantization and continuity; this avoids redundancy and ensures consistency across the work.

2.2 Per-Step Testability Theorem

Theorem 2.1 (Per-Step Testability $\Leftrightarrow$ Information Loss at Resolution ε)**.** *Let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ be a continuous update map on compact metric spaces, and let the current belief state and incoming measurement be denoted $(\Theta_t, z^{\text{in}}_{t+1})$, with next belief $\Theta_{t+1} = U(\Theta_t, z^{\text{in}}_{t+1})$. Fix a finite resolution parameter $\varepsilon > 0$ and let all relevant quantities be quantized/coarsened to precision ε as in Chapter 3 (so that indistinguishability is defined via ε-balls).*

Define the per-step certification language L_{step} to consist of observed coarse outputs (heads) for which a given tail predicate remains certifiable in the sense of a finite witness for the next-step behavior. Then:

1. *If U exhibits nontrivial information loss at resolution ε (e.g., positive mutual information drop or, equivalently in the finite quantized picture, a fiber over some coarse output has size at least two), then $L_{\text{step}} \in \mathbf{NP}$.*

2. *If U is lossless at resolution ε (i.e., no collapse: the coarse observed next state uniquely determines the quantized preimage up to the precision relevant to the predicate), then certifying the tail behavior requires solving a universally quantified continuation, rendering L_{step} not in $\mathbf{NP}$ in general (e.g., Π_2^0-hard as in the lossless reduction from* TOTAL*).*

Proof. We reduce the statement to the framework and theorems of Book I, Chapter 5 (the "information loss $\Leftrightarrow$ testability" theorem for discrete/quantized channels) by explicitly matching the Book II objects to the earlier setup.

Setup alignment with Book I. In Book I the measurement channel was described as a deterministic map from a discrete (or finitely quantized) input to an output, and testability was characterized via the existence of short witnesses when the channel was non-invertible at that resolution. Here:

- The pair $(\Theta_t, z_{t+1}^{\text{in}})$ plays the role of the joint input in Book I; the next belief $\Theta_{t+1} = U(\Theta_t, z_{t+1}^{\text{in}})$ is the channel output.

- We impose a finite-resolution quantization of $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ by covering them with ε-balls and identifying any two points within the same ball as indistinguishable. This produces a finite alphabet model identical in spirit to the discrete setting of Book I. All entropy and mutual information statements are taken with respect to this induced finite partition.

- The tail predicate under consideration is assumed to be robust to ε-perturbations (as in the admissibility definition), so that behavior simulated from any representative of an ε-ball yields the same verdict.

Thus the per-step channel

$$(\Theta_t, z_{t+1}^{\text{in}}) \longmapsto \Theta_{t+1}$$

with quantization at scale ε fits exactly into the Book I template: the possibility of certification (membership of a "good head") is governed by whether the channel is information-losing (non-injective or collapsing) at that resolution.

(1) Sufficiency: Information loss $\Rightarrow$ NP certification. Assume the channel loses information at resolution ε: in the finite quantized model, there exists at least one coarse output (i.e., ε-ball of Θ_{t+1}) whose preimage contains two distinct coarse inputs (θ, z^{in}) and

$(\theta', z^{\mathrm{in}\prime})$ that are separated at the input level but collide after applying U. Equivalently, the mutual information

$$I\big((\Theta_t, z^{\mathrm{in}}_{t+1}); \Theta_{t+1}\big) < H\big(\Theta_t, z^{\mathrm{in}}_{t+1}\big)$$

in the quantized/discrete formulation, so there is a positive information drop. By the sufficiency direction of the Book I theorem, such non-invertibility provides the combinatorial structure necessary to build a short witness:

- The witness consists of (a) the coarse label of Θ_{t+1} (the observed output cell) and (b) an index identifying which of the indistinguishable cluster/fiber-members the true input belonged to (i.e., a representative within the preimage fiber up to the finite ambiguity). This mirrors the "coarse-label plus fiber index" witness of Book I, Chapter 5.
- Because the tail predicate is ε-robust, simulating the continuation from any state within the ε-ball consistent with the witness yields the same predicate value.
- Continuity and compactness guarantee that the number of such clusters is bounded (metric entropy controls the size), so the witness length is $O(\log(1/\varepsilon))$ (or polynomial in the relevant input size) and can be encoded efficiently.
- The verifier can, in deterministic polynomial time, (i) reconstruct a representative state from the witness, (ii) simulate U and subsequent dynamics to check consistency with the observed head, and (iii) evaluate the tail predicate on the simulated continuation, accepting exactly the true instances.

Therefore $L_{\mathrm{step}} \in \mathbf{NP}$ under information loss, completing the sufficiency direction by invoking the corresponding constructive proof from Book I with the quantization layer prepended.

(2) Necessity: Losslessness $\Rightarrow$ hardness beyond NP. Suppose instead that at resolution ε no information is lost: the coarse output Θ_{t+1} uniquely determines the coarse input $(\Theta_t, z^{\mathrm{in}}_{t+1})$ relevant for the tail predicate to within the allowable ambiguity. Equivalently, the finite-resolution channel is injective on the support of interest (or has zero mutual information drop), so there is no nontrivial fiber collapse to exploit. Any candidate finite witness that attempts to certify the tail behavior must, in effect, resolve universal quantification over all possible consistent continuations: since the future behavior is uniquely tied to the current state, and no collapse provides a "shortcut" ambiguity to witness, the verifier cannot reduce the certificate to an existential check without solving a statement of the form "for all possible continuations the tail predicate holds," which is of the Π^0_2-type in the arithmetic hierarchy. This is exactly the situation analyzed in the hardness direction in Book I (e.g., the reduction from TOTAL in the lossless model): the certification problem becomes at least Π^0_2-hard, and hence cannot be placed in **NP** unless unlikely collapses occur in the complexity hierarchy.

Conclusion. The two directions together yield the claimed equivalence at finite resolution: per-step testability (membership of the certification language in **NP**) holds precisely when there is per-step information loss at that resolution. The formal machinery and reductions are inherited from Book I after aligning the update map and quantization, as done above. $\square$

Remark. The theorem separates the numerical (information-theoretic) condition $\Delta_I^{(\varepsilon)} > 0$ from the algorithmic outcome (**NP**-certifiability), emphasizing that the presence of ambiguity at finite resolution is the enabling resource, while its absence yields intractability in the worst case.

2.3 Uniform Control of Finite-Resolution Information Loss under Strict Contraction

The goal of this subsection is to make precise the quantitative statement that on a region where U acts as a uniform strict contraction, any observable per-step information loss at finite resolution can be driven to zero by refining the resolution. This provides the "numerical upper bound" ingredient used in Theorem 3.1.

Setup 2.1. *Let $(\mathcal{M}, d)$ be a compact metric space and $\mathcal{Z}^{\text{in}}$ another compact metric space. Suppose the update map $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ satisfies the following on a subset $R \subseteq \mathcal{M}$:*

1. ***Uniform strict contraction:*** *There exists $\rho \in [0, 1)$ such that for all $\theta, \theta' \in R$ and all $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$,*
$$d\big(U(\theta, z^{\text{in}}),\, U(\theta', z^{\text{in}})\big) \le \rho \cdot d(\theta, \theta').$$

2. ***Uniform continuity in the other arguments:*** *For each fixed $\theta \in R$, the map $z^{\text{in}} \mapsto U(\theta, z^{\text{in}})$ is uniformly continuous on $\mathcal{Z}^{\text{in}}$ (automatic from continuity on compact sets).*

Proposition 2.1 (Vanishing Finite-Resolution Information Loss under Contraction)**.** *Under the above setup, consider the finite-resolution mutual information drop $\Delta_I^{(\varepsilon)}$ computed by quantizing Θ_t and z_{t+1}^{in} to precision $\varepsilon > 0$ (as in Chapter 3), and letting $\Theta_{t+1} = U(\Theta_t, z_{t+1}^{\text{in}})$. Then*
$$\lim_{\varepsilon \to 0} \Delta_I^{(\varepsilon)} = 0.$$
In other words, for every $\delta > 0$ there exists $\varepsilon_0 > 0$ such that for all $0 < \varepsilon < \varepsilon_0$, the observable mutual information drop is less than δ.

Proof. We sketch the argument in metric-entropy terms; the same intuition carries over to any finite quantization model used to define $\Delta_I^{(\varepsilon)}$.

Fix $\delta > 0$. Because U is a strict contraction on R in the θ-coordinate with factor $\rho < 1$, the image under U of any ε-ball in the θ-variable is contained in a ball of radius at most $\rho\varepsilon$. Thus the number of ε-balls needed to cover the range of U (for fixed z^{in}) is no more than the number needed to cover R with ε/ρ-balls, which in turn is controlled by its metric entropy.

More concretely: let $N_\theta(\varepsilon)$ be the minimal number of balls of radius ε needed to cover R. Then the effective uncertainty in Θ_t at resolution ε gets compressed to uncertainty of size $\rho\varepsilon$ in Θ_{t+1} (holding z^{in} fixed), while uncertainty coming from z^{in} enters only through a uniformly continuous map and hence can be controlled separately by refining its quantization. Therefore, the joint uncertainty in $(\Theta_t, z_{t+1}^{\text{in}})$ that survives into Θ_{t+1} shrinks as $\varepsilon \to 0$.

In information-theoretic terms, this means the mutual information $I((\Theta_t, z_{t+1}^{\text{in}}); \Theta_{t+1})$ approaches the total available information (since the contraction makes the mapping from

Θ_t to Θ_{t+1} increasingly deterministic at fine scales), so the drop

$$\Delta_I^{(\varepsilon)} \;=\; H(\Theta_t, z_{t+1}^{\text{in}}) - I((\Theta_t, z_{t+1}^{\text{in}}); \Theta_{t+1})$$

tends to zero. Formally, because all spaces are compact and U is uniformly continuous jointly, for any $\eta > 0$ there exists ε_0 so that for $\varepsilon < \varepsilon_0$ the distortion induced by quantization and contraction makes the conditional entropy $H((\Theta_t, z_{t+1}^{\text{in}}) \mid \Theta_{t+1}) < \eta$, implying $\Delta_I^{(\varepsilon)} < \eta$.

Thus $\lim_{\varepsilon \to 0} \Delta_I^{(\varepsilon)} = 0$, proving the proposition. □

Remark. The key ingredients are the uniform shrinkage of uncertainty in the belief-state under contraction and the compactness/uniform continuity that prevent pathological decoupling between input quantization and output behavior. No global injectivity is assumed or needed for this vanishing bound; only the local contraction property suffices to squeeze finite-resolution loss arbitrarily small.

3 Double-Bound Theorem: Resolution, Contraction, and Semi-Decidability

Overview. In systems with deterministic dynamics that are *injective* in the infinite-precision limit, one might nevertheless observe an *effective loss of distinguishability* at finite resolution due to contraction: nearby initial states produce trajectories that become hard to tell apart over short observation windows. This creates an apparent paradox: contraction both hides differences (numerical vanishing) and, simultaneously, enables stabilization that supports algorithmic refinement of the hidden state.

The **Double-Bound Theorem** resolves this tension. It articulates two coexisting phenomena restricted to a contraction basin:

1. A *numerical bound* (vanishing distinguishability): finite-resolution uncertainty is exponentially damped by contraction, so short prefixes carry little information about initial perturbations.

2. An *algorithmic semi-decision/refinement bound*: despite the temporary indistinguishability, there exists a procedure that progressively certifies ever-finer approximations of the hidden state (digit-by-digit) and detects genuine inconsistency in finite time.

Together, these imply that while finite observation depth and coarse resolution mask differences (making the system appear non-injective), appropriate refinement in depth and precision recovers full information (restoring effective injectivity), reconciling the "information-loss" intuition with the underlying injective structure.

3.1 Definitions

Definition 3.1 (Numerical Error Propagation)**.** Let θ_0 be the true initial state and $\widehat{\theta}_0$ a candidate with $d(\theta_0, \widehat{\theta}_0) \le \delta_0$. Define the propagated states $\theta_t, \widehat{\theta}_t$ by forward iteration under

U (using the same inputs). Then, by contraction,

$$d(\theta_t, \widehat{\theta}_t) \leq \rho^t \delta_0.$$

We refer to $\delta_t := d(\theta_t, \widehat{\theta}_t)$ as the numerical error at time t.

Definition 3.2 (Finite-Window Observation Prefix)**.** Let $Z^{(T)}(\theta)$ denote the length-T observation prefix generated deterministically from initial state θ, possibly after passing through an output/emission map. The finite-resolution observed prefix $\widetilde{Z}^{(T)}$ is a noisy version of $Z^{(T)}(\theta)$ with per-coordinate distortion at most $O(\varepsilon)$; more precisely, the distribution of $\widetilde{Z}^{(T)}$ conditioned on θ is within total variation $O(\varepsilon)$ of the ideal prefix.

Definition 3.3 (Distinguishability and Mutual Information)**.** For random initial state Θ taking values in a finite support $\mathcal{S} \subseteq B_r(\theta^*)$, the mutual information $I(\Theta; \widetilde{Z}^{(T)})$ measures how much of the randomness in Θ is retained in the finite-resolution length-T prefix $\widetilde{Z}^{(T)}$. We define the *information gap* as $H(\Theta) - I(\Theta; \widetilde{Z}^{(T)})$.

Definition 3.4 (Refinement / Semi-Decision Procedure)**.** A refinement (semi-decision) procedure is an algorithm that, given a sequence of candidate approximations $\widehat{\theta}^{(k)}$ with target precisions $\varepsilon_k \to 0$, and observing longer trajectory prefixes as needed, (i) never accepts an incorrect approximation beyond its claimed precision, and (ii) eventually confirms correct refinements by producing certified digits or approximations. It rejects any candidate whose observable trajectory deviates beyond the tolerance implied by contraction at finite depth.

3.2 Intermediate Lemmas

Lemma 3.1 (Numerical Bound / Error Vanishing)**.** *Under the local contraction assumption in $B_r(\theta^*)$, if the true initial state is θ_0 and the candidate is $\widehat{\theta}_0$ with $d(\theta_0, \widehat{\theta}_0) \leq \delta_0$, then after t iterations the propagated discrepancy satisfies*

$$d(\theta_t, \widehat{\theta}_t) \leq \rho^t \delta_0.$$

Thus small initial uncertainty is exponentially damped in time.

Lemma 3.2 (Finite-Window Mutual Information Attenuation)**.** *Let Θ be a random variable supported on a finite set $\mathcal{S} \subset B_r(\theta^*)$ and let $\widetilde{Z}^{(T)}$ be the finite-resolution observed length-T prefix generated from Θ. Then there exists a function $g(\rho^T, \varepsilon) = O((\rho^T \varepsilon)^2)$ such that*

$$I(\Theta; \widetilde{Z}^{(T)}) \geq H(\Theta) - g(\rho^T, \varepsilon).$$

Equivalently, the information gap $H(\Theta) - I(\Theta; \widetilde{Z}^{(T)})$ vanishes as $\rho^T \varepsilon \to 0$.

Lemma 3.3 (Resolution-to-Injectivity)**.** *For any two distinct states $\theta, \theta' \in B_r(\theta^*)$ and any target distinguishability $\gamma > 0$, there exist finite window length T and resolution $\varepsilon > 0$ such that the corresponding finite-resolution prefixes $\widetilde{Z}^{(T)}(\theta)$ and $\widetilde{Z}^{(T)}(\theta')$ differ with total variation at least γ. In particular, as $\rho^T \varepsilon \to 0$, the finite-resolution observation map becomes effectively injective on the support.*

3.3 Main Double-Bound Theorem

Theorem 3.1 (Double-Bound Theorem)**.** *Let the system satisfy the assumptions of Section 2 restricted to the contraction basin $B_r(\theta^*)$. Let Θ be a random initial state supported inside the basin, and let $\widetilde{Z}^{(T)}$ be the finite-resolution observed trajectory prefix of length T. Then:*

1. ***Numerical Bound:*** *Any finite-resolution uncertainty in the initial condition is exponentially suppressed by contraction: small differences between candidate initials are difficult to distinguish from short prefixes. Quantitatively, if two initials differ by δ_0, their induced observation distributions over length T prefixes satisfy divergence bounds of order $O((\rho^T\delta_0)^2)$, so the mutual information about the initial state satisfies*

$$I(\Theta; \widetilde{Z}^{(T)}) \geq H(\Theta) - g(\rho^T, \varepsilon)$$

   with $g(\rho^T, \varepsilon) = O((\rho^T\varepsilon)^2)$.

2. ***Algorithmic Semi-Decision Bound:*** *There exists a refinement procedure which, given a candidate $\widehat{\theta}_0$ at resolution ε, observes a prefix of sufficient length T (determined by ρ, ε, and desired accuracy) and either:*

   - *Certifies the approximation to precision ε (emitting the corresponding digits) if the observed trajectory is consistent within the contracted tolerance, or*
   - *Rejects in finite time if the trajectory deviates beyond the allowed error induced by contraction.*

3. ***Coexistence / Recovery:*** *Despite the temporary attenuation of distinguishability (item 1), injectivity ensures that by appropriate simultaneous refinement of window length T and resolution ε (so that $\rho^T\varepsilon \to 0$) the mutual information $I(\Theta; \widetilde{Z}^{(T)}) \to H(\Theta)$. Hence the refinement procedure converges to the true initial state, achieving effective injectivity in the limit.*

3.4 Algorithmic Corollary: Digit-by-Digit Refinement

Corollary 3.1 (Refinement Procedure)**.** *Under the same assumptions as Theorem 3.1, there is an explicit algorithm that produces a sequence of certified approximations $\widehat{\theta}^{(k)}$ converging to the true θ as follows:*

1. *Fix a decreasing sequence of resolutions $\varepsilon_1 > \varepsilon_2 > \cdots \to 0$.*
2. *For each k, determine a prefix length T_k large enough that $\rho^{T_k}\varepsilon_{k-1} < \varepsilon_k$ (with ε_0 an initial coarse bound).*
3. *Using the candidate certificate $\widehat{\theta}^{(k)}$, simulate the predicted trajectory and compare to the observed finite-resolution prefix $\widetilde{Z}^{(T_k)}$.*
4. *If consistency holds within the contracted tolerance, emit the k-th chunk/digits corresponding to precision ε_k; otherwise reject the certificate.*

This procedure never certifies an incorrect digit (soundness) and, when run on data generated inside the contraction basin with a correct underlying state, produces a convergent sequence $\widehat{\theta}^{(k)} \to \theta$ (completeness).

3.5 Proofs

Proof of Lemma 3.1. Immediate by induction on t: assume $d(\theta_t, \widehat{\theta}_t) \leq \rho^t \delta_0$. Then using contraction,

$$d(\theta_{t+1}, \widehat{\theta}_{t+1}) = d(U(\theta_t, z_t), U(\widehat{\theta}_t, z_t)) \leq \rho\, d(\theta_t, \widehat{\theta}_t) \leq \rho^{t+1}\delta_0.$$

□

Proof of Lemma 3.2. By contraction and the assumed finite-resolution blur, the conditional distributions of $\widetilde{Z}^{(T)}$ given different initials are pairwise close in total variation: for any θ, θ',

$$\mathrm{TV}\big(\mathcal{L}(\widetilde{Z}^{(T)} \mid \theta), \mathcal{L}(\widetilde{Z}^{(T)} \mid \theta')\big) = O(\rho^T d(\theta, \theta') + \varepsilon).$$

Then standard bounds (e.g., Pinsker's inequality and Fano-type arguments) imply that the conditional entropy $H(\Theta \mid \widetilde{Z}^{(T)})$ is upper-bounded by $O((\rho^T\varepsilon)^2)$, yielding

$$I(\Theta; \widetilde{Z}^{(T)}) = H(\Theta) - H(\Theta \mid \widetilde{Z}^{(T)}) \geq H(\Theta) - g(\rho^T, \varepsilon)$$

with $g = O((\rho^T\varepsilon)^2)$. □

Proof of Lemma 3.3. Given distinct θ, θ', injectivity ensures their infinite trajectories differ. Contraction implies that at finite depth T, their trajectory prefixes remain separated by at least $\Delta_T = d(Z^{(T)}(\theta), Z^{(T)}(\theta')) - O(\rho^T d(\theta, \theta'))$. Choosing T large enough so that $\rho^T d(\theta, \theta')$ is small relative to the intrinsic divergence, and taking ε sufficiently small to not obscure this separation, the finite-resolution observed prefixes differ by at least γ. Thus effective injectivity obtains as $\rho^T\varepsilon \to 0$. □

Proof of Theorem 3.1. Item (1) is Lemma 3.2. Item (2) is achieved by using the known contraction rate: the refinement procedure chooses T_k so that previous uncertainty is damped below the next target precision and tests consistency; failure is detectable because any deviation beyond $\rho^{T_k}\varepsilon_{k-1}$ will eventually manifest in the observed prefix and trigger rejection. Item (3) follows from Lemma 3.3: as $T \to \infty$ and $\varepsilon \to 0$ in tandem, the mutual information gap vanishes, so the procedure converges to the true underlying state. □

Proof of Corollary 3.1. Directly follows from Theorem 3.1: the scheduling T_k ensures that resolution refinement is supported by sufficient trajectory depth to distinguish, and the contraction-based tolerance guarantees no false-positive certification. Completeness is due to injectivity and vanishing information gap. □

3.6 Extensions and Remarks

Noisy or Approximate Dynamics. If the update map U or observation process includes stochastic perturbations, the numerical bound degrades gracefully: one replaces deterministic contraction with expected contraction or probabilistic mixing rates, and the refinement procedure incorporates confidence intervals. The mutual information recovery still holds under appropriate ergodicity/stability conditions.

Connection to Guard-Band and Certification Standards. The Double-Bound mechanism underpins why guard-band predicates coupled with local contraction yield robust certificates: the numerical bound explains why local uncertainties can be controlled, and the refinement (algorithmic) bound supplies the procedure to verify consistency and progressively tighten the guard-band witness.

Information-Theoretic Interpretation. The mutual information statement (Lemma 3.2) makes precise the claim that finite-window observations temporarily "forget" small differences due to contraction, yet the underlying injectivity ensures no permanent loss: information is recoverable in the limit.

Resolution → Injectivity Limit. As a corollary of Lemma 3.3, for any fixed finite support prior P_Θ,

$$\lim_{\substack{T\to\infty \\ \varepsilon\to 0}} I(\Theta; \widetilde{Z}^{(T)}) = H(\Theta),$$

i.e., the system behaves as if injective under arbitrarily fine observation and sufficient depth.

3.7 Epistemic Losslessness: Definition and Formalization

The double-bound phenomenon motivates a distinction between *algebraic* information preservation (injectivity in the infinite-precision ideal) and what is observable and certifiable in practice under contraction and finite resolution. We call the latter property *epistemic losslessness*.

Definition 3.5 (Epistemic Losslessness)**.** Let $U : \mathcal{M} \times \mathcal{Z}^{\text{in}} \to \mathcal{M}$ be the update map, and suppose $R \subseteq \mathcal{M}$ is a region on which U satisfies a local strict contraction with factor $\rho \in [0, 1)$. Fix a random initial state Θ supported in R, and for each resolution parameter $\varepsilon > 0$ and prefix length T, let $\widetilde{Z}_\varepsilon^{(T)}$ denote the finite-resolution observed prefix (as in Section earlier) arising from Θ. Define the *information gap*

$$\Delta_I^{(T,\varepsilon)} := H(\Theta) - I\big(\Theta; \widetilde{Z}_\varepsilon^{(T)}\big),$$

and write $\Delta_I^{(\varepsilon)} := \inf_T \Delta_I^{(T,\varepsilon)}$ for the best achievable residual uncertainty at resolution ε.

We say U is *epistemically lossless* on R if the following two properties hold:

1. **Vanishing Information Gap:** For every $\delta > 0$ there exists $\varepsilon > 0$ sufficiently small and a corresponding window length T such that $\Delta_I^{(T,\varepsilon)} < \delta$. Equivalently, $\lim_{\substack{\varepsilon\to 0 \\ T\to\infty}} \Delta_I^{(T,\varepsilon)} = 0$.
2. **Refinable Semi-Decidable Certification:** Restricted to R, the per-step certification language admits a semi-decision (refinement) procedure that, given a candidate initial state approximation and observing progressively longer prefixes at appropriately increasing precision, (i) never incorrectly accepts a false claim beyond its certified precision, and (ii) eventually confirms true instances by producing arbitrarily fine certified approximations. In particular, true statements are recursively enumerable in the infinite-precision limit.

Discussion. Epistemic losslessness does *not* assert that U is globally injective in a naive finite-resolution sense or that there is zero ambiguity at any fixed finite depth. Rather, it formalizes that:

- Observable ambiguity can be driven below any user-specified threshold by refining resolution and/or extending observation length; the apparent loss induced by contraction is transient and vanishes in the limit (numerical vanishing).
- Verification of a true hypothesis proceeds via a controlled refinement process (semi-decision) rather than requiring an oracle for a non-recursive predicate; legitimate claims can be corroborated by finite, progressively stronger evidence.

Consequences.

1. In epistemically lossless regions, finite-resolution witnesses become *effectively unique* up to any desired precision: once the information gap is below a threshold, further observations produce diminishing *epistemically significant* new information.
2. The transition from regimes with a strictly positive lower bound on $\Delta_I^{(\varepsilon)}$ (persistent information ambiguity) to epistemically lossless regimes corresponds to moving from settings where only nonrefinable certificates exist to those where one can iteratively refine and certify the hidden state—i.e., from coarse **NP**-style certifiability to a refinement-based semi-decidable confirmation paradigm.

Chapter 5: The Minimal Condition for Invariant Basins: Local Contraction

Dmytro Surdu

In the previous chapter, we proved that per-step testability is equivalent to per-step information loss. However, this is not sufficient to certify claims about the infinite future. This chapter introduces the central structure required for tail certification: the ***Attractor Basin****. We will carefully build the axioms required for a useful basin theory and then prove the* ***Minimality Theorem****. This crucial result establishes that a uniform strict contraction on the basin is the necessary and sufficient condition for basin invariance, given reachability and a well-behaved boundary. This theorem provides the rigorous justification for the primary tool of our theory.*

1 The Logical Gap and the Need for a Basin

We have established that we can trust the computed belief-state Θ_t at any finite time t. But consider a typical tail predicate:

> P_{tail}: "For all future times $k > n$, the system's belief-state Θ_k will remain within a pre-defined safe set $\mathcal{S}$."

Even if we know that our belief-state's uncertainty shrinks at every step, this provides no finite point at which we can stop and declare the infinite-tail property proven. An asymptotic guarantee is not a finite proof. To bridge this gap, we need to identify a "point of no return"—a region in the belief-space which, once entered, guarantees the tail predicate holds forever. This is the Attractor Basin.

2 Axioms for a Usable Attractor Basin

For an attractor basin $\mathcal{B}$ to be useful for certification, it must be reachable in a bounded time and it must be invariant (once entered, it is never left). We begin by formalizing these properties as axioms.

Definition 2.1 (Pointwise Reachability (weak R))**.** A closed set $\mathcal{B} \subseteq \mathcal{M}$ satisfies $(\mathbf{R}_{\text{pt}})$ if for every initial $\Theta_0 \in \mathcal{M}$ and every admissible input stream $(z_1, z_2, \dots)$ there exists a (possibly instance–dependent) $T \geq 0$ such that $U^T(\Theta_0; z_{1:T}) \in \mathcal{B}$.

Remark 2.1 (Role of the Lyapunov border)**.** Write $\mathcal{B} = \{\theta : V(\theta) \leq r\}$ as in Axiom L. Let $\mathring{\mathcal{B}} = \{\theta : V(\theta) < r\}$ denote the strict interior. By Axiom L, if $V(\Theta) = r$ then for *every* z^{in} we have $V(U(\Theta, z^{\text{in}})) < r$, i.e. $U(\Theta, z^{\text{in}}) \in \mathring{\mathcal{B}}$. Consequently, any trajectory that enters $\mathcal{B}$ at some time t is guaranteed to be in $\mathring{\mathcal{B}}$ by time $t+1$, regardless of the next input.

Definition 2.2 (Axiom L: Lyapunov Border Condition)**.** A closed set $\mathcal{B} = \{\theta \in \mathcal{M} \mid V(\theta) \leq r\}$ defined by a continuous function $V : \mathcal{M} \to [0, \infty)$ and a constant $r > 0$ satisfies **Axiom L** if every boundary point is mapped strictly inside the set on the next update. Formally:

$$\forall \Theta \in \mathcal{M} \text{ with } V(\Theta) = r, \quad \forall z^{\text{in}} \in \mathcal{Z}^{\text{in}} : \quad V(U(\Theta, z^{\text{in}})) < r$$

This condition prevents trajectories from "touching" the boundary of the basin and immediately leaving. We now define the final property of interest.

Definition 2.3 (Axiom C: Local Contraction on $\mathcal{B}$)**.** The update map U satisfies **Axiom C** on a set $\mathcal{B}$ if there exists a constant $\rho \in [0, 1)$ such that for all $\Theta, \Theta' \in \mathcal{B}$ and all $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$:

$$d(U(\Theta, z^{\text{in}}), U(\Theta', z^{\text{in}})) \leq \rho \cdot d(\Theta, \Theta')$$

3 The Minimality Theorem

The main result of this chapter is that, given reachability and a well-behaved border, local contraction is not just a sufficient condition for invariance, but also a necessary one.

Theorem 3.1 (Minimality of Local Contraction)**.** *Let $(\mathcal{M}, d)$ be a compact metric space, and let U and a closed set $\mathcal{B} \subseteq \mathcal{M}$ be given. Assume that Axiom R (Uniform Reachability) and Axiom L (Lyapunov Border) hold for $\mathcal{B}$. Then, the set $\mathcal{B}$ is invariant under U if and only if U satisfies Axiom C (Local Contraction on $\mathcal{B}$).*

Before proving the theorem in both directions, we lift the weak reachability to the uniform reachability with the following lemma.

Lemma 3.1 (Uniformization from compactness and the basin axioms)**.** *Assume:*

1. (C)*: $\mathcal{M}$ and $\mathcal{Z}^{\text{in}}$ are compact metric spaces, and U is continuous;*
2. (R_{weak}) *(Def. 2.1);*
3. (L) *(Lyapunov border for $\mathcal{B} = \{V \leq r\}$);*
4. (C_{basin}) *(local contraction on $\mathcal{B}$; not used in the proof below but part of the basin axiom pack).*

Then there exists a finite $T_{\max}$ such that for every $\Theta_0 \in \mathcal{M}$ and every admissible input stream $\mathbf{z}$*,*

$$U^{T_{\max}}(\Theta_0; z_{1:T_{\max}}) \in \mathcal{B}.$$

In particular, (R) *(uniform reachability) holds and is* derived *from* (C)*,* (R_{weak})*, and* (L)*.*

Proof. Equip the infinite input space $\mathcal{Z}^{\text{in}\,\mathbb{N}}$ with the product topology. Since $\mathcal{Z}^{\text{in}}$ is compact metric by (C), Tychonoff's theorem yields that $\mathcal{Z}^{\text{in}\,\mathbb{N}}$ is compact; being a countable product of compact metrizable spaces, it is again compact metric.

For each $t \in \mathbb{N}$, define the continuous map

$$F_t : \ \mathcal{M} \times \mathcal{Z}^{\text{in}\,\mathbb{N}} \ \longrightarrow \ \mathcal{M}, \qquad F_t(\Theta_0, \mathbf{z}) \ := \ U^t(\Theta_0; z_{1:t}),$$

where U^t is the t–fold composition of the continuous update U (continuity follows by induction on t and continuity of the coordinate projection $\mathbf{z} \mapsto z_{1:t}$).

Let $\mathring{\mathcal{B}} = \{V < r\}$ be the interior of $\mathcal{B}$. For each $t \geq 1$, set

$$\mathcal{O}_t \ := \ F_t^{-1}(\mathring{\mathcal{B}}) \ \subseteq \ \mathcal{M} \times \mathcal{Z}^{\text{in}\,\mathbb{N}}.$$

Because F_t is continuous and $\mathring{\mathcal{B}}$ is open, each $\mathcal{O}_t$ is open.

Now fix any pair $(\Theta_0, \mathbf{z})$. By (R_{weak}), there exists T with $U^T(\Theta_0; z_{1:T}) \in \mathcal{B}$. By Remark 2.1, regardless of z_{T+1} we then have $U^{T+1}(\Theta_0; z_{1:T+1}) \in \mathring{\mathcal{B}}$, i.e. $(\Theta_0, \mathbf{z}) \in \mathcal{O}_{T+1}$. Therefore the open sets $\{\mathcal{O}_t\}_{t \geq 1}$ form an open cover of the compact space $\mathcal{M} \times \mathcal{Z}^{\text{in}\,\mathbb{N}}$.

By compactness, there exists a finite subcover

$$\mathcal{M} \times \mathcal{Z}^{\text{in}\,\mathbb{N}} \ = \ \bigcup_{i=1}^{k} \mathcal{O}_{t_i}.$$

Let $T_{\max} = \max\{t_1, \ldots, t_k\}$. Then for every $(\Theta_0, \mathbf{z})$ there is an index i with $(\Theta_0, \mathbf{z}) \in \mathcal{O}_{t_i}$, hence $U^{t_i}(\Theta_0; z_{1:t_i}) \in \mathring{\mathcal{B}} \subseteq \mathcal{B}$, and therefore $U^{T_{\max}}(\Theta_0; z_{1:T_{\max}}) \in \mathcal{B}$ as well (because $\mathcal{B}$ is forward–invariant under the $V < r$ guard by (L), and in any case $t_i \leq T_{\max}$ gives entry by time $T_{\max}$). Thus a uniform bound exists. □

3.1 Proof of Sufficiency (⇐)

We prove that if a map is a strict contraction on a set with a Lyapunov border, it must be invariant on that set.

Recall the setting: $\mathcal{M}$ is a compact metric space, $\mathcal{B} = \{\theta \in \mathcal{M} \mid V(\theta) \leq r\}$ is closed, and we assume

- **Axiom L (Lyapunov Border)**: $\forall\, \Theta \in \mathcal{M}$ with $V(\Theta) = r$, $\forall\, z^{\text{in}} \in \mathcal{Z}^{\text{in}} : \ V\big(U(\Theta, z^{\text{in}})\big) < r$.
- **Axiom C (Local Contraction)**: $\exists\, \rho \in [0, 1)$ such that $\forall\, \Theta, \Theta' \in \mathcal{B}$, $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$:

$$d\big(U(\Theta, z^{\text{in}}),\, U(\Theta', z^{\text{in}})\big) \ \leq \ \rho\, d(\Theta, \Theta') \ .$$

We will show that these two imply

$$\forall\, \Theta \in \mathcal{B},\, \forall\, z^{\text{in}} \in \mathcal{Z}^{\text{in}} : \quad U(\Theta, z^{\text{in}}) \in \mathcal{B},$$

i.e. $\mathcal{B}$ is invariant.

Lemma 3.2 (Boundary Preservation)**.** *Under Axiom L, every boundary point is sent strictly into the interior:*

$$\forall\, \Theta \text{ with } V(\Theta) = r,\ \forall\, z^{in}: \quad V\big(U(\Theta, z^{in})\big) < r\ .$$

Proof. This is exactly Axiom L. □

Lemma 3.3 (Interior Preservation via Quantitative Contraction)**.** *Under Axiom L we have a strictly positive "Lyapunov gap"*

$$\delta \;=\; r \;-\; \max_{\substack{\Theta\in\mathcal{M}\\ V(\Theta)=r}}\ \max_{z\in\mathcal{Z}^{\mathrm{in}}} V\big(U(\Theta,z)\big) \;>\; 0.$$

If, in addition, the map U satisfies Axiom C with a contraction constant ρ such that

$$L_V\,\rho\,\mathrm{diam}(\mathcal{B}) \;<\; \delta,$$

where L_V is a Lipschitz constant of V on $\mathcal{B}$, then every interior point is sent into the strict interior:

$$\forall\, \Theta \text{ with } V(\Theta) < r,\ \forall\, z \in \mathcal{Z}^{\mathrm{in}}: \quad V\big(U(\Theta, z)\big) < r.$$

Proof. Since $V \circ U(\cdot, z)$ is continuous on the compact boundary $\{\Theta : V(\Theta) = r\}$, it attains the maximum $\max_{V(\Theta)=r} V(U(\Theta, z))$. Define $\delta = r - \max_{V(\Theta)=r} V(U(\Theta, z)) > 0$. Also, by Axiom C, for any boundary point Θ_b and any interior point Θ,

$$d\big(U(\Theta, z),\, U(\Theta_b, z)\big) \;\le\; \rho\, d(\Theta, \Theta_b) \;\le\; \rho\,\mathrm{diam}(\mathcal{B}).$$

Since V is Lipschitz on $\mathcal{B}$ with constant L_V, we have

$$V\big(U(\Theta, z)\big) \;\le\; V\big(U(\Theta_b, z)\big) \;+\; L_V\, d\big(U(\Theta, z),\, U(\Theta_b, z)\big) \;\le\; (r - \delta) \;+\; L_V\,\rho\,\mathrm{diam}(\mathcal{B}).$$

By the hypothesis $L_V\,\rho\,\mathrm{diam}(\mathcal{B}) < \delta$, the right-hand side is strictly less than r. Hence $V(U(\Theta, z)) < r$, as required. □

Proof of Sufficiency. **2. Boundary-gap and quantitative contraction.** By Axiom L, the boundary of $\mathcal{B}$ is sent strictly inside, yielding

$$\delta \;=\; r \;-\; \max_{V(\Theta)=r}\ \max_{z} V\big(U(\Theta,z)\big) \;>\; 0.$$

Invoke the Lipschitz constant L_V of V on $\mathcal{B}$. We require $\rho < 1$ to satisfy

$$L_V\,\rho\,\mathrm{diam}(\mathcal{B}) \;<\; \delta,$$

which is exactly the strengthened Axiom C. Then by Lemma 3.3, both boundary and interior points map into $\{V < r\}$. Hence for every $\Theta \in \mathcal{B}$ and $z \in \mathcal{Z}^{\mathrm{in}}$, $V(U(\Theta, z)) < r$, proving invariance of $\mathcal{B}$.

□

Thus, by Lemma 3.1, (R) holds (uniform reachability is derived).

3.2 Proof of Necessity ($\Rightarrow$)

We recall the setting and axioms:

- **Uniform Reachability (Axiom R).** There exists a finite $T_{\max}$ such that for every initial state $\Theta_0 \in \mathcal{M}$ and every admissible measurement stream $(z_1, z_2, \dots)$, the trajectory $\Theta_t = U^t(\Theta_0; z_1, \dots, z_t)$ enters $\mathcal{B}$ by time $T_{\max}$.
- **Lyapunov Border (Axiom L).** The set $\mathcal{B} = \{\theta \in \mathcal{M} : V(\theta) \le r\}$ is closed, and
$$\forall\, \Theta \text{ with } V(\Theta) = r,\ \forall\, z \in \mathcal{Z}: \quad V\big(U(\Theta, z)\big) < r.$$
- **Invariance.** We assume $\mathcal{B}$ is invariant: $\Theta \in \mathcal{B},\ z \in \mathcal{Z} \implies U(\Theta, z) \in \mathcal{B}$.

We must show that U is a uniform strict contraction on $\mathcal{B}$:

$$\exists\, \rho \in [0, 1) \quad \text{such that} \quad \forall\, \Theta, \Theta' \in \mathcal{B},\ z \in \mathcal{Z}:\ d\big(U(\Theta, z), U(\Theta', z)\big) \le \rho\, d(\Theta, \Theta').$$

Lemma 3.4 (Supremum Attainment)**.** *Let $(\mathcal{B}, d)$ and $(\mathcal{Z}, \rho)$ be compact metric spaces, and let*
$$U : \mathcal{B} \times \mathcal{Z} \longrightarrow \mathcal{B}$$
be continuous. For any $\delta > 0$, define the closed subset
$$D_\delta \;=\; \{\, (\Theta, \Theta', z) \in \mathcal{B} \times \mathcal{B} \times \mathcal{Z} : d(\Theta, \Theta') \ge \delta\}.$$
Define
$$F : D_\delta \longrightarrow \mathbb{R}\,, \qquad F(\Theta, \Theta', z) = \frac{d\big(U(\Theta, z),\, U(\Theta', z)\big)}{d(\Theta, \Theta')}.$$
Then F is continuous on the compact set D_δ, and hence there exists
$$(\bar{\Theta}, \bar{\Theta}', \bar{z}) \;\in\; D_\delta$$
such that
$$F(\bar{\Theta}, \bar{\Theta}', \bar{z}) \;=\; \max_{(\Theta, \Theta', z) \in D_\delta} F(\Theta, \Theta', z) \;=\; \sup_{(\Theta, \Theta', z) \in D_\delta} F(\Theta, \Theta', z).$$
In particular, the supremum of F on D_δ is attained.

Proof. 1. **Compactness of D_δ.** Since $\mathcal{B}$ and $\mathcal{Z}$ are compact, their product $\mathcal{B} \times \mathcal{B} \times \mathcal{Z}$ is compact. The function
$$(\Theta, \Theta', z) \longmapsto d(\Theta, \Theta')$$
is continuous, so $\{\, (\Theta, \Theta', z) : d(\Theta, \Theta') \ge \delta\}$ is the preimage of the closed set $[\delta, \infty)$ and thus closed. A closed subset of a compact space is compact, so D_δ is compact.

2. **Continuity of F.**
   - The map $(\Theta, \Theta', z) \mapsto d(U(\Theta, z), U(\Theta', z))$ is continuous as a composition of the continuous functions U and the metric d.

- The denominator $(\Theta, \Theta', z) \mapsto d(\Theta, \Theta')$ is continuous and bounded below by $\delta > 0$ on D_δ, so it never vanishes there.
- Hence F, as the quotient of two continuous functions with nonzero denominator, is continuous on D_δ.

3. **Application of the Extreme Value Theorem.** Because F is continuous on the compact set D_δ, it must attain both its maximum and minimum. Therefore there exists a point $(\bar{\Theta}, \bar{\Theta}', \bar{z}) \in D_\delta$ where F achieves its supremum, as claimed.

□

Lemma 3.5 (Near–Non-Contraction)**.** *Let $(\mathcal{B}, d)$ and $(\mathcal{Z}, \rho)$ be compact metric spaces, let*

$$U : \mathcal{B} \times \mathcal{Z} \to \mathcal{B}$$

be continuous, and define

$$F\big(\Theta, \Theta', z\big) = \frac{d\big(U(\Theta, z),\, U(\Theta', z)\big)}{d(\Theta, \Theta')} \quad \text{on} \quad D = \{(\Theta, \Theta', z) : \Theta \neq \Theta'\}.$$

Set

$$\rho \;=\; \sup_{(\Theta, \Theta', z) \in D} F(\Theta, \Theta', z).$$

Then for every $\varepsilon > 0$ there exists $\delta > 0$ and a point $(\bar{\Theta}, \bar{\Theta}', \bar{z}) \in \mathcal{B} \times \mathcal{B} \times \mathcal{Z}$ with $d(\bar{\Theta}, \bar{\Theta}') \geq \delta$ such that

$$F(\bar{\Theta}, \bar{\Theta}', \bar{z}) \;\geq\; \rho \;-\; \varepsilon.$$

Proof. For each $\delta > 0$ define the closed set

$$D_\delta = \{\, (\Theta, \Theta', z) \in \mathcal{B} \times \mathcal{B} \times \mathcal{Z} : d(\Theta, \Theta') \geq \delta\}.$$

Since $\bigcup_{\delta > 0} D_\delta = D$, we have

$$\rho = \sup_{D} F = \sup_{\delta > 0} \sup_{D_\delta} F.$$

Note that $\delta \mapsto \sup_{D_\delta} F$ is a non-increasing function and $\lim_{\delta \to 0^+} \sup_{D_\delta} F = \rho$. Hence given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\sup_{D_\delta} F \;>\; \rho - \varepsilon.$$

By Lemma 3.4, F attains its supremum on the compact set D_δ. Thus there exists $(\bar{\Theta}, \bar{\Theta}', \bar{z}) \in D_\delta$ with

$$F(\bar{\Theta}, \bar{\Theta}', \bar{z}) = \sup_{D_\delta} F > \rho - \varepsilon,$$

and by construction $d(\bar{\Theta}, \bar{\Theta}') \geq \delta > 0$. This $(\bar{\Theta}, \bar{\Theta}', \bar{z})$ is the desired off-diagonal witness. □

Lemma 3.6 (Slow-Convergence vs. Bounded Reachability, refined)**.** *Let* $(\mathcal{B}, d)$ *and* $(\mathcal{Z}, \rho)$ *be compact metric spaces, and let*

$$U : \mathcal{X} \times \mathcal{Z} \to \mathcal{X}$$

be continuous on the ambient space $\mathcal{X} \supset \mathcal{B}$*. For any* $\delta > 0$ *define the compact set*

$$D_\delta^* = \big\{(\Theta, \Theta', z) \in \big(\mathcal{X} \setminus \mathcal{B}\big) \times \partial\mathcal{B} \times \mathcal{Z} : d(\Theta, \Theta') \geq \delta\big\}.$$

Set

$$\rho_\delta^* = \sup_{(\Theta,\Theta',z)\in D_\delta^*} \frac{d\big(U(\Theta, z), U(\Theta', z)\big)}{d(\Theta, \Theta')} .$$

Then for every $\varepsilon > 0$ *there exists* $\delta > 0$ *and a witness* $(\Theta_0, \Theta_0', z^*) \in D_\delta^*$ *with*

$$\frac{d\big(U(\Theta_0, z^*),\, U(\Theta_0', z^*)\big)}{d(\Theta_0, \Theta_0')} \;>\; \rho_\delta^* - \varepsilon,$$

so in particular $d_0 = d(\Theta_0, \Theta_0') > 0$ *and* $\Theta_0' \in \partial\mathcal{B}$*.*

Proof. By simple induction,

$$d\big(\Theta_t, \mathcal{B}\big) \;\geq\; (1-\varepsilon)\, d\big(\Theta_{t-1}, \mathcal{B}\big) \;\geq\; \cdots \;\geq\; (1-\varepsilon)^t\, d(\Theta_0, \mathcal{B}).$$

□

Proof of Necessity. Assume $\mathcal{B}$ is invariant but U is not a strict contraction, so its minimal Lipschitz constant on $\mathcal{B}$ satisfies $L_\mathcal{B} \geq 1$. By Lemma 3.5, for any small $\varepsilon > 0$ there exist $\theta_0, \theta_0' \in \mathcal{B}$ and $z^* \in \mathcal{Z}^{\text{in}}$ with

$$d(\theta_0, \theta_0') \;=\; d_0 > 0, \quad d\big(U(\theta_0, z^*), U(\theta_0', z^*)\big) \;\geq\; (1-\varepsilon)\, d_0.$$

Now fix the adversarial input stream which repeats z^* at every step. Starting from θ_0, invariance ensures we stay in $\mathcal{B}$. By Lemma **??**, each update satisfies

$$d(\theta_t, \mathcal{B}) \;\geq\; (1-\varepsilon)^t\, d_0 \;>\; 0$$

for all $t \leq T_{\max}$, provided $(1-\varepsilon)^{T_{\max}} d_0 > 0$. Thus no trajectory can reach $\mathcal{B}$ within $T_{\max}$ steps, contradicting Uniform Reachability (Axiom R). Hence $L_\mathcal{B} < 1$, i.e. U is a strict contraction on $\mathcal{B}$. □

4 Corollaries: The Two Design Patterns

The Minimality Theorem shows that local contraction on a basin is essential. This leads to two primary ways to construct a certifiable system.

Corollary 4.1 (Global Contraction)**.** *If the update map* U *is a global uniform strict contraction on the entire space* $\mathcal{M}$*, then one can choose the basin to be* $\mathcal{B} = \mathcal{M}$*. Axioms R and L hold automatically.*

Proof. If U is a contraction on $\mathcal{M}$, then $\mathcal{B} = \mathcal{M}$ is trivially invariant. Reachability is also trivial since every trajectory starts in $\mathcal{M}$ ($T_{\max} = 0$). The Lyapunov condition can be satisfied by taking $V(\theta) = 0$ for all θ. This is the simplest certifiable structure. □

Corollary 4.2 (Local Contraction + Guard-Band)**.** *If one only assumes (1) Local Contraction on a proper subset $\mathcal{B} \subset \mathcal{M}$ and (2) Uniform Reachability into $\mathcal{B}$, this is equivalent to imposing a "measurement guard-band" on the input observations z_t^{in}.*

Theorem 4.1. *Suppose*

1. ***Reachability*** *(R) to some closed $\mathcal{B}$.*
2. *A* ***Guard-Band*** *condition on input observations: there is a* ***proper*** *subset $\mathbb{R} \subsetneq \mathcal{Z}^{in}$ (the "acceptance region") such that*

$$\forall\, \theta \in \mathcal{B},\ \forall\, z^{in} \in \mathbb{R}: \quad U(\theta, z^{in}) \ \in\ \mathcal{B}.$$

(But for inputs outside $\mathbb{R}$ we make no claim.)

Then $\mathcal{B}$ is forward-invariant once the stream enters it, and since $\mathbb{R} \neq \mathcal{Z}^{in}$ one also has $\mathcal{B} \neq \mathcal{M}$.

Proof. To prove the theorem, we note that:

1. **Invariance under Guard-Band.** Let $\theta \in \mathcal{B}$. By reachability, after some finite t, the true input measurement z_t^{in} must lie in the guard-band $\mathbb{R}$ (otherwise you could never force entry to $\mathcal{B}$). Once $\theta_t \in \mathcal{B}$ and $z_{t+1}^{\text{in}} \in \mathbb{R}$, the Guard-Band rule says $U(\theta_t, z_{t+1}^{\text{in}}) \in \mathcal{B}$. Inductively every subsequent update remains in $\mathcal{B}$.
2. $\mathcal{B} \neq \mathcal{M}$. If $\mathcal{B} = \mathcal{M}$, then invariance would hold for *all* $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$, not just those in $\mathbb{R}$. But by hypothesis the basin only "locks in" under $z^{\text{in}} \in \mathbb{R}$, so there must exist at least one $\bar{z}^{\text{in}} \notin \mathbb{R}$ and at least one $\bar{\theta} \in \mathcal{B}$ such that we make no invariance claim for $U(\bar{\theta}, \bar{z}^{\text{in}})$. Ergo $\mathcal{B} \subsetneq \mathcal{M}$.

Finally, by the Minimality Theorem the combination R + invariance (from the Guard-Band) + L (automatic once you enlarge $\mathcal{B}$ to be closed) *implies* a local contraction constant $\rho < 1$ on $\mathcal{B}$. □

Chapter 6: The Attractor Basin Equivalence Theorem

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With the full machinery of belief-states, per-step testability, and the minimal axiom of contraction in place, we are now ready to prove the central result of our Type B theory. This chapter will demonstrate that the existence of a reachable Attractor Basin is the necessary and sufficient condition for a tail predicate to be deterministically certifiable. This equivalence provides a complete characterization of testability, showing that the abstract logical requirement of having a short proof corresponds perfectly to the concrete structural property of having a stable, invariant "safe region" in the system's belief-space.

1 The Main Equivalence

In Chapter 5, we established that a useful theory of tail certification requires an Attractor Basin defined by the properties of contraction, safety, and reachability. We now elevate this concept from a useful tool to the cornerstone of a fundamental equivalence.

Before starting, we clarify the statement we want to certify.

Axiom 1.1 (Emission Safety)**.** *Let*

$$S_{\text{out}} = \{\, z^{out} \in \mathcal{Z}^{out} : \textit{the output } z^{out} \textit{ satisfies the terminal condition}\}.$$

Then

$$\mathcal{B} \;\subseteq\; \{\Theta \in \mathcal{M} : E(\Theta) \in S_{\text{out}}\}.$$

If invariance is assumed, this immediately extends to

$$\forall\, t \geq 0,\ \forall\, \Theta \in \mathcal{B},\ \forall\, z^{in}_{1:t} \in (\mathcal{Z}^{in})^t,\ E\big(U^t(\Theta, z^{in}_{1:t})\big) \in S_{\text{out}}.$$

Axiom 1.2 (Polytime Verifier Axiom)**.** *There exist a polynomial $q(\cdot)$ and a deterministic polynomial–time verifier V for P_{tail} with the following properties.*

(V1) ***Polytime verifier.*** *On input size n (the length of the observed output head), V runs in time $q(n)$.*

(V2) ***Belief–state encoding and evaluability.*** *There is a fixed computable encoding* $\text{enc} : \mathcal{M} \to \{0,1\}^\star$ *and a decoding/evaluation procedure such that for any $\varepsilon > 0$ and $\Theta \in \mathcal{M}$, from* $\text{enc}_\varepsilon(\Theta)$ *of length $O(\log(1/\varepsilon))$ the verifier can evaluate $E(\Theta)$ and $U(\Theta, \cdot)$ to accuracy ε in time* $\text{poly}(\log(1/\varepsilon))$*. Consequently, along any T–step capture certificate, the total state–side encoding length and the evaluation cost are* $\text{poly}(T)$*.*

(V3) ***Honest–read on yes–instances.*** *On every accepting run (i.e., when the provided multi–step capture certificate is correct), V reads the entirety of that certificate, including the binary encoding of T and all per–step subcertificates.*

In other words, our certifiable statement about measurement S_{out} must be true on $\mathcal{B}$.

Definition 1.1 (Admissible Safe Predicate)**.** Define $P_{\text{tail}} = S_{\text{out}}$, a predicate on the infinite output stream in $(\mathcal{Z}^{\text{out}})^{\omega}$.

Then the following theorem holds:

Theorem 1.1 (The Attractor Basin Equivalence)**.** *A tail predicate P_{tail} over the output stream of a deterministic Type B model is deterministically certifiable by a polynomial-time scheme if and only if there exists a compact set $\mathcal{B} \subset \mathcal{M}$ (the universal–tail basin) such that:*

(R) ***Reachability:*** $\forall\, \Theta_0 \in \mathcal{M},\ \exists\, T \in \mathbb{N}$ *and input head* $z^{\text{in}}_{1:T}$ *with* $U^T(\Theta_0, z^{\text{in}}_{1:T}) \in \mathcal{B}$.

(I) ***Invariance (over inputs):*** $\forall\, \Theta \in \mathcal{B},\ \forall\, z^{\text{in}} \in \mathcal{Z}^{\text{in}},\ U(\Theta, z^{\text{in}}) \in \mathcal{B}$.

(S) ***Safety (state–only emission):*** $\forall\, \Theta \in \mathcal{B},\ E(\Theta) \in S_{\text{out}}$, *hence every tail from* Θ *satisfies* P_{tail}.

(C) ***Contraction on $\mathcal{B}$:*** *There exists* $\alpha \in (0,1)$ *with*

$$d\big(U(\Theta, z^{\text{in}}), U(\Theta', z^{\text{in}})\big) \le \alpha\, d(\Theta, \Theta') \quad \forall\, \Theta, \Theta' \in \mathcal{B},\ \forall\, z^{\text{in}} \in \mathcal{Z}^{\text{in}}.$$

Moreover, in the $\Rightarrow$ direction we first obtain (R), (I), (S); then (C) follows from the Minimality Theorem of Chapter 5 by invoking its Lyapunov–Border axiom for $\mathcal{B}$ (equivalently, one may prove (C) directly on the compact invariant set). The certificate can be taken as a capture certificate $(T, \widehat{\Theta}, z^{\text{in}}_{1:T})$ *attesting both reachability* $U^T(\Theta_0, z^{\text{in}}_{1:T}) = \widehat{\Theta} \in \mathcal{B}$ *and the fact that every tail from* $\widehat{\Theta}$ *satisfies* P_{tail}.

This theorem is the direct analogue of the Guard-Band Equivalence for Type A systems. It establishes that the only way to certify an infinite-tail property is to demonstrate that the system's state has entered a region from which failure is impossible. We prove each direction of the equivalence separately.

2 Proof of Necessity ($\Rightarrow$)

We start with the harder and the more profound direction. We assume a certification system exists and deduce that its underlying structure must be equivalent to an attractor basin. This proof "reverse-engineers" testability.

Lemma 2.1 (Verifier-Normalization)**.** *Let V be any deterministic Turing machine which, on input*

$$\big(z_{1:n},\, w\big), \quad z_{1:n} \in (\mathcal{Z}^{\text{out}})^n, \quad w \in \{0,1\}^{\le p(n)},$$

runs in time $\text{poly}(n)$ *and accepts exactly those pairs in its language. Then there exists a deterministic Turing machine V' running in time* $\text{poly}(n)$ *such that:*

1. *On any input* $\big(z_{1:n},\, w'\big)$*,* V' *inspects only the first* n *symbols of* $z_{1:n}$ *and only the first* ℓ *bits of* w'*, where* $\ell \le p(n)$ *is declared in a self-delimiting prefix of* w'*.*

2. V' *accepts* $\big(z_{1:n}, w'\big)$ *if and only if* w' *correctly encodes a witness* w *of length* $\ell \le p(n)$ *and* V *accepts* $(z_{1:n}, w)$*.*

Hence V' *accepts exactly the same set of pairs* $(z_{1:n}, w)$ *as* V*, while guaranteed to ignore any extra input or padding.*

Proof. We construct V' as follows. On input $\big(z_{1:n},\, w'\big)$:

1. Parse the declared lengths.

- Read from the start of w' a self-delimiting encoding of an integer ℓ. If decoding fails or $\ell > p(n)$, *reject*.
- Read the next ℓ bits of w' as the *actual* witness w_{actual}. If w' has fewer than ℓ bits after the prefix, *reject*.

2. Enforce head-length bound.

- Check that the input head is at least n symbols long. (By hypothesis it is exactly n.)
- Record the fact that no simulation step may read beyond symbol n of the head or beyond bit ℓ of the witness.

3. Simulate V under guard.

- Initialize a standard simulation of V on input $(z_{1:n}, w_{\text{actual}})$, but *interpose* the following rule: Before each simulated transition, if V's tape-head on the input tape is positioned beyond cell n, or if V's tape-head on the witness tape is positioned beyond cell ℓ, then *halt and reject* immediately.
- Otherwise, perform exactly the transition that V would perform.

4. Accept or reject.

- If the guarded simulation of V enters an accepting state, then *accept*.
- If it enters a rejecting state (or was forced to reject by an out-of-bounds access), then *reject*.

Time bound. The self-delimiting decoding of ℓ takes poly(n) time, and the guarded simulation of V adds only a constant-factor overhead to V's own poly(n) runtime. Hence V' runs in poly(n) time.

Correctness. On any "honest" input $(z_{1:n}, w)$ with $|w| \le p(n)$, we can form $w' = \langle\, |w|, w \rangle$. Then V' parses $\ell = |w|$, simulates V exactly on the first n symbols and first ℓ bits, and cannot be forced to reject by an out-of-bounds check. Thus V' accepts if and only if V accepts.

Therefore V' meets both requirements of the lemma. □

Lemma 2.2 (Suffix Certification)**.** *Let V be a deterministic verifier which, on input*

$$(z_{1:n}, w), \quad |w| \le p(n),$$

runs in time $\mathrm{poly}(n)$ *and accepts exactly those pairs for which the tail predicate holds. Suppose*

$$V(z_{1:n}, w) = \textit{accept}.$$

By Lemma 2.1, we may assume without loss of generality that V inspects only the first n symbols of its head and the first $p(n+m)$ bits of its witness. Define

$$w' \;=\; w \parallel 0^{p(n+m)-|w|},$$

so that $|w'| = p(n+m)$. Then

$$V\big(z_{1:n+m},\, w'\big) \;=\; \textit{accept}.$$

Proof. By Lemma 2.1, replace V with the normalized verifier which cannot read beyond position n of the head or bit $p(n+m)$ of the witness. Since the original run $V(z_{1:n}, w)$ only depends on those first n symbols and $|w|$ bits, feeding $(z_{1:n+m}, w')$ reproduces the same computation and accepts. □

Lemma 2.3 (Suffix Universality)**.** *Let $V'(V)$ be a normalized deterministic verifier for the tail-predicate P_{tail} on infinite measurement streams, and let $p : \mathbb{N} \to \mathbb{N}$ bound the witness length. Assume:*

- *(i) Soundness* *If $V'(z_{1:N}, w)$ accepts, then any infinite extension $z_{1:\infty} \supset z_{1:N}$ satisfies $P_{\mathrm{tail}}(z_{1:\infty})$.*
- *(ii) Completeness If $P_{\mathrm{tail}}(z_{1:\infty})$ holds, then for every n there exists w with $|w| \le p(n)$ such that $V'(z_{1:n}, w)$ accepts.*
- *(iii) Tail–Invariance If $P_{\mathrm{tail}}(z_{1:\infty})$ holds, then for every $1 \le k < n$, the suffix $z_{k+1:\infty}$ also satisfies P_{tail}.*

Then for any $N \ge 1$, if there is a witness w with $|w| \le p(N)$ such that

$$V'\big(z_{1:N},\, w\big) \;=\; \text{“}\textit{accept,}\text{”}$$

then for every $1 \le k < N$ there exists a witness w' with $|w'| \le p(N-k)$ such that

$$V'\big(z_{k+1:N},\, w'\big) \;=\; \text{“}\textit{accept.}\text{”}$$

Proof. Fix $N \ge 1$ and suppose w of length $\le p(N)$ certifies the head $z_{1:N}$:

$$V'\big(z_{1:N}, w\big) \;=\; \text{accept}.$$

By (i) Soundness, any infinite continuation $z_{1:\infty}$ extending $z_{1:N}$ satisfies

$$P_{\mathrm{tail}}(z_{1:\infty}).$$

Then by (iii) Tail–Invariance, for each $1 \le k < N$ the suffix $z_{k+1:\infty}$ also satisfies the tail predicate:

$$P_{\text{tail}}(z_{k+1:\infty}).$$

By (ii) Completeness, for each such suffix of finite length $N-k$ there must exist a witness w' with $|w'| \le p(N-k)$ such that

$$V'\big(z_{k+1:N},\, w'\big) \;=\; \text{accept}.$$

Since k was arbitrary in $\{1, \ldots, N-1\}$, this constructs the required certificate for every suffix of the original head. □

Necessity: Certifiability ⇒ Existence of a Basin. Assume that the tail predicate P_{tail} over the output stream in $(\mathcal{Z}^{\text{out}})^\omega$ is deterministically certifiable. Concretely, there exists a polynomial $p(n)$ and a deterministic, polynomial-time verifier

$$V^* : \big((\mathcal{Z}^{\text{out}})^n \times \mathcal{M}\big) \times \{0,1\}^{\le p(n)} \;\longrightarrow\; \{\text{accept}, \text{reject}\},$$

with the property that for any initial belief-state Θ_0, any finite output measurement prefix $z^{\text{out}}_{1:n}$, and its corresponding (unobserved) input prefix $z^{\text{in}}_{1:n}$, there is a certificate triple

$$(n, \widehat{\Theta}, w), \quad \widehat{\Theta} \in \mathcal{M}, \quad |w| \le p(n),$$

such that

$$V^*\big((\, z^{\text{out}}_{1:n}, \widehat{\Theta}),\, w\big) = \text{accept} \quad \Longleftrightarrow \quad \begin{cases} U^n(\Theta_0, z^{\text{in}}_{1:n}) = \widehat{\Theta}, \\ P_{\text{tail}}\big(z^{\text{out}}_{n+1:\infty}(\widehat{\Theta})\big) \text{ holds}, \end{cases}$$

i.e. V^* accepts exactly when the claimed state $\widehat{\Theta}$ is reached in n steps and the infinite continuation from $\widehat{\Theta}$ satisfies the tail predicate. The witness w must contain sufficient information (such as the input history $z^{\text{in}}_{1:n}$) to allow the verifier to check the path.

We must show that this assumption implies the existence of a set $\mathcal{B} \subseteq \mathcal{M}$ that satisfies the three properties of a refined Attractor Basin: Safety, Contraction (implying Invariance), and Reachability.

Step 1: Constructing the Basin Candidate $\mathcal{B}$. Let us define our candidate set $\mathcal{B}$ as the set of all belief-states Θ for which a valid, accepting witness exists, assuming an empty observation history (i.e., at time $n = 0$).

Definition 2.1 (Basin Candidate and Universal–Tail Basin)**.** Let $(\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E)$ be a deterministic Type B model and let (V^*, p) be a sound and complete polytime tail–certifier for P_{tail}. Define:

(i) Basin Candidate (eventual–entry set).

$$\mathcal{B}_{\text{cand}} \;=\; \Big\{\Theta_0 \in \mathcal{M} :\; \exists\, T \in \mathbb{N},\, \widehat{\Theta} \in \mathcal{M},\, w \in \{0,1\}^{\le p(T)} \text{ s.t. } V^*((z^{\text{out}}_{1:T}(\Theta_0), \widehat{\Theta}), w) = \text{accept}\Big\}.$$

Intuitively, these are the states from which *some* finite prefix admits a valid capture certificate.

(ii) Universal–Tail Basin (already–safe core).

$$\mathcal{B}^\star \;=\; \Big\{\Theta \in \mathcal{M} :\; \forall\, z^{\text{in}}_{1:\infty} \in (\mathcal{Z}^{\text{in}})^\omega,\; P_{\text{tail}}\big(\text{Out}(\Theta; z^{\text{in}}_{1:\infty})\big) \text{ holds}\Big\}.$$

This is the set of states that are safe *now* and remain safe for every admissible input continuation.

In what follows we write $\mathcal{B} := \mathcal{B}^\star$ for the invariant safe region used in ABET; $\mathcal{B}_{\text{cand}}$ is only used in reachability arguments.

Then, the following lemma holds:

Lemma 2.4 (Closedness and Compactness of the Basin Candidate)**.** *Assume*

- $\mathcal{M}$ *and* $\mathcal{Z}$ *are compact metric spaces,*
- $U : \mathcal{M} \times \mathcal{Z} \to \mathcal{M}$ *is continuous.*

Then the basin candidate

$$\mathcal{B}_{\text{cand}} = \{\Theta_0 \in \mathcal{M} : \exists\, z_{1:T} \in \mathcal{Z}^T,\, U^T(\Theta_0, z_{1:T}) = \Theta_T\}$$

is a compact (hence closed) subset of $\mathcal{M}$*.*

Here $\mathcal{B}_{\text{cand}}$ denotes the candidate set of states that can reach the fixed Θ_T; it is not assumed to satisfy safety.

Proof. Define the map

$$F : \mathcal{M} \times \mathcal{Z}^T \;\longrightarrow\; \mathcal{M}, \qquad F(\Theta_0, z_{1:T}) = U^T(\Theta_0, z_{1:T}).$$

1. **Continuity.** Since U is continuous, its T-fold composition F is continuous on the product $\mathcal{M} \times \mathcal{Z}^T$.
2. **Preimage is closed.** The singleton $\{\Theta_T\} \subset \mathcal{M}$ is closed. By continuity of F, the set
$$F^{-1}(\{\Theta_T\}) = \{\, (\Theta_0, z_{1:T}) : F(\Theta_0, z_{1:T}) = \Theta_T\}$$
is closed in the compact space $\mathcal{M} \times \mathcal{Z}^T$. Hence $F^{-1}(\{\Theta_T\})$ is compact.
3. **Projection preserves compactness.** $\mathcal{B}_{\text{cand}}$ is exactly the image of $F^{-1}(\{\Theta_T\})$ under the projection $\pi : \mathcal{M} \times \mathcal{Z}^T \to \mathcal{M}$, $\pi(\Theta_0, z_{1:T}) = \Theta_0$. Since π is continuous, $\pi\big(F^{-1}(\{\Theta_T\})\big) = \mathcal{B}_{\text{cand}}$ is compact.
4. **Compact $\implies$ Closed.** In any metric space, compactness implies closedness. Therefore $\mathcal{B}_{\text{cand}}$ is also closed in $\mathcal{M}$.

□

Step 2 (Safety $\Rightarrow$ Invariance). Let $\mathcal{B}^\star$ be the set constructed in Step 1. We now show that $\mathcal{B}^\star$ is *invariant* under the update map U from Definition 2.1, Chapter 3. Fix any $\Theta \in \mathcal{B}^\star$ and any admissible measurement $z^{\text{in}} \in \mathcal{Z}^{\text{in}}$ (Definition 1.2, Chapter 3). By the *Safety* property of $\mathcal{B}^\star$, every trajectory starting in Θ satisfies the tail predicate; in particular, the concatenated infinite sequence formed by (Θ, z^{in}) followed by any admissible continuation is also safe. Tail-invariance of the predicate (Axiom 1.2) implies that if the tail of a trajectory satisfies the predicate, then so does the entire trajectory starting from the updated belief-state $U(\Theta, z^{\text{in}})$. Thus $U(\Theta, z^{\text{in}}) \in \mathcal{B}^\star$. Since Θ and z^{in} were arbitrary, we conclude $\mathcal{B}^\star$ is forward-invariant under U, establishing (I).

Step 3: The Impasse of Uniform Reachability. Our next goal is to prove that the basin candidate $\mathcal{B}$ is reachable from any initial state $\Theta_0 \in \mathcal{M}$. While the assumption of NP-certifiability guarantees that for any given Θ_0 there exists *some* finite time n at which a certificate can be found, it does not guarantee a *uniform* time bound $T_{\max}$ that works for all initial states. Without such a uniform bound, the basin may not be reachable in any practical or certifiable sense.

This reveals a critical gap in our theory: the verifier V has been treated as a pure algorithm, but to guarantee uniform reachability, we must analyze its properties in the continuous, topological setting of our state-space. We must now resolve this by formalizing the nature of the verifier.

Resolving the Gap: The Certification Standard

To bridge the gap between per-state certifiability and uniform reachability, we must ensure that the verifier's acceptance is not a "knife-edge" phenomenon. A small perturbation in the measurement stream should not invalidate a correct witness. This leads to a notion of robustness.

A First Attempt: The Robust Verifier Axiom. One could simply posit an axiom that the verifier itself is robust:

> *Axiom (Robust Verifier): The acceptance region of the verifier V is an open set in the product topology of outputs and witnesses.*

While this solves the problem mathematically, it is ontologically unsatisfying. It forces the verifier V, a purely epistemic and computational object, to adopt properties of the physical, continuous world, blurring a useful distinction.

An Ontologically Pure Solution: The Certification Standard. A more elegant solution is to keep the verifier as a pure algorithm and introduce a separate, purely epistemic object that codifies the required robustness. This object is the **Certification Standard**. It acts as a formal contract specifying the tolerance of any certification.

Definition 2.2 (Certification Standard)**.** A **Certification Standard**, denoted $\mathcal{S}$, is a family of guard-band radii $\{\varepsilon_n\}_{n=1}^{\infty}$, where each $\varepsilon_n \in (0, 1]$ if an **acceptance by the verifier V on**

output $(z_{1:n}^{\mathbf{out}}, w)$ **under** $\mathcal{S}$ makes V also accepting on the entire ε_n-ball around $(z_{1:n}^{\text{out}}, w)$ in the **product metric** on

$$\mathcal{Z}^{\text{out n}} \times \mathcal{W}$$

built from:

1. **The output-space metric** a metric $d_{\mathcal{Z}^{\text{out}}}$ on $\mathcal{Z}^{\text{out}}$ built from a usual metric on a compact metric space $\mathcal{Z}^{\text{out}}$ as a standard **sup** (or l_∞) metric on length-n tuples:

$$d_{\mathcal{Z}^n}\big((z_1, \ldots, z_n), (z_1', \ldots, z_n')\big) = \max_{1 \le i \le n} \; d_{\mathcal{Z}^{\text{out}}}(z_i, z_i').$$

2. **The witness-space metric** built from witnesses as finite bit-strings of bounded (polynomial) length, which equips $\mathcal{W}$ with the **discrete metric**

$$d_{\mathcal{W}}(w, w') = \begin{cases} 0, & w = w', \\ 1, & w \neq w'. \end{cases}$$

3. **As the product metric** on $\mathcal{Z}^{\text{out n}} \times \mathcal{W}$ as the maximum of the two:

$$d\big((z_{1:n}, w), (z_{1:n}', w')\big) = \max\Big\{ d_{\mathcal{Z}^n}(z_{1:n}, z_{1:n}'),\; d_{\mathcal{W}}(w, w') \Big\}.$$

   so a set $W_n \subseteq \mathcal{Z}^{\text{out n}} \times \mathcal{W}$ is **open** in this metric precisely if, whenever $(z^*, w^*) \in W_n$, there is an $\varepsilon > 0$ so that

$$d\big((z_{1:n}, w), (z^*, w^*)\big) < \varepsilon \implies (z_{1:n}, w) \in W_n.$$

   and, therefore, for each n, the acceptance relation

$$W_n = \{(z_{1:n}, w) \mid V(z_{1:n}, w) = \text{accept}\}$$

   is open in $\mathcal{Z}^{\text{out n}} \times \mathcal{W}$. Such an acceptance relation is said to be **robustly certified by** $\mathcal{S}$.

This leads to a more complete definition of the object of our study.

Definition 2.3 (The Testable Measurement)**.** A non-trivial **Testable Measurement** is a sextuple $\mathfrak{M} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, E, U, \mathcal{S})$, where $(\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}})$ are the physical measurement spaces, update and emit maps (E, U) are the physical dynamics and $\mathcal{S}$ is the epistemic Standard under which Testable Measurement properties are certified.

Armed with this, we can now rigorously prove the Uniform Reachability property.

Lemma 2.5 (Uniform Reachability via Compactness)**.** *Assume $\mathcal{M}$, $\mathcal{Z}^{\text{in}}$, $\mathcal{Z}^{\text{out}}$ are compact metric spaces, U and E are continuous, and (V, p) is a sound and complete polytime tail–certifier that is* robust *under the Certification Standard $\mathcal{S}$ (acceptance sets are open in the product metric for each fixed witness length bound). For each $n \geq 1$, define*

$$A_n := \Big\{\Theta_0 \in \mathcal{M} : \ \exists\, w, \ |w| \leq p(n), \ V\big(z^{\text{out}}_{1:n}(\Theta_0), w\big) = \text{accept under } \mathcal{S}\Big\}.$$

Then:

1. $\bigcup_{n\geq 1} A_n = \mathcal{M}$.
2. *Each A_n is open in $\mathcal{M}$.*
3. *By compactness, there exists $T_{\max} \in \mathbb{N}$ with $\bigcup_{n\leq T_{\max}} A_n = \mathcal{M}$.*

Proof. (1) Completeness of V yields, for each Θ_0, some n and witness w with acceptance; hence $\Theta_0 \in A_n$.

(2) Fix n. For each binary string w with $|w| \leq p(n)$, define

$$W_{n,w} := \{z_{1:n} \in (\mathcal{Z}^{\text{out}})^n : V(z_{1:n}, w) = \text{accept under } \mathcal{S}\}.$$

By robustness of $\mathcal{S}$, each $W_{n,w}$ is open in $(\mathcal{Z}^{\text{out}})^n$. Let $\Phi_n : \mathcal{M} \to (\mathcal{Z}^{\text{out}})^n$, $\Phi_n(\Theta_0) = z^{\text{out}}_{1:n}(\Theta_0)$; continuity of E, U implies Φ_n is continuous. Then

$$A_n \ = \bigcup_{|w|\leq p(n)} \Phi_n^{-1}(W_{n,w}),$$

a finite union of open sets; hence A_n is open.

(3) $\{A_n\}_{n\geq 1}$ is an open cover of compact $\mathcal{M}$; extract a finite subcover and define $T_{\max}$ as its maximum index. □

Step 3 (Resumed): Proving Reachability. The *Uniform Reachability via Compactness* lemma, proven above, establishes that our basin candidate $\mathcal{B}$ is reachable from any initial state within the uniform time bound $T_{\max}$. This closes the logical gap.

Step 4 (Lyapunov–border via one–step distance). Let $\mathcal{B}^\star \subseteq \mathcal{M}$ be the universal–tail safe set constructed in Steps 1–3; by definition it is closed and forward–invariant under all inputs. Define the one–step distance functional

$$L(\Theta) \ := \ \sup_{z\in\mathcal{Z}} \ \mathrm{dist}\big(U(\Theta, z), \, \mathcal{M} \setminus \mathcal{B}^\star\big), \qquad \Theta \in \mathcal{B}^\star,$$

where $\mathrm{dist}(x, A) := \inf\{d(x, a) : a \in A\}$. We prove that L is a bona fide Lyapunov–border function on $\mathcal{B}^\star$, i.e. L is continuous, $L(\Theta) = 0$ iff $\Theta \in \partial\mathcal{B}^\star$, and there exists $\alpha \in (0, 1)$ such that

$$\boxed{L(\Theta) \ \leq \ \alpha\, \mathrm{dist}\big(\Theta, \, \mathcal{M} \setminus \mathcal{B}^\star\big) \quad \forall\, \Theta \in \mathcal{B}^\star.} \qquad (\star)$$

Continuity. Since U is continuous and $\mathcal{Z}$ is compact, the map $\Theta \mapsto \mathrm{dist}\big(U(\Theta, z), \, \mathcal{M}\setminus\mathcal{B}^\star\big)$ is continuous for each fixed z, and the supremum over a compact index set preserves upper

semicontinuity; joint Lipschitzness of U on the compact domain implies full continuity of L on $\mathcal{B}^\star$.

Zero set. If $\Theta \in \partial\mathcal{B}^\star$, then $U(\Theta, z) \in \mathcal{B}^\star$ for all z (forward–invariance) and, by the definition of boundary, $\operatorname{dist}\big(U(\Theta, z), \mathcal{M}\setminus\mathcal{B}^\star\big) = 0$ for arbitrarily close choices of z–perturbations; hence $L(\Theta) = 0$. Conversely, if $L(\Theta) = 0$ then along some sequence z_n we have $\operatorname{dist}\big(U(\Theta, z_n), \mathcal{M}\setminus \mathcal{B}^\star\big) \to 0$, which—by forward–invariance—forces Θ to be a limit point of $\mathcal{M} \setminus \mathcal{B}^\star$; hence $\Theta \in \partial\mathcal{B}^\star$.

Strict vanishing only on the boundary. If $\Theta \in (\mathcal{B}^\star)^\circ$, compactness and invariance give a neighborhood $B(\Theta, r)$ with $U(B(\Theta, r), \mathcal{Z}) \subset (\mathcal{B}^\star)^\circ$; thus $L(\Theta) > 0$.

One–step proportional inwardness. Fix any $\Theta \in \mathcal{B}^\star$ and write $\delta(\Theta) := \operatorname{dist}\big(\Theta, \mathcal{M} \setminus \mathcal{B}^\star\big)$. Let L_U denote a global Lipschitz constant of U in Θ on $\mathcal{B}^\star \times \mathcal{Z}$ (exists by compactness). By Step 3 (Certification Standard) applied at one step and the continuity of $E \circ U$, there exists an *output robustness radius* $\rho > 0$ such that: if $d\big(E(U(\Theta, z)),\, E(U(\Theta', z))\big) \le \rho$ for all z, then safety persists one step (the acceptance at length 1 is open). *By compactness of $\mathcal{B}^\star$ (Definition 1.2, Chapter 3) and openness of the acceptance set under the Certification Standard (Definition 2.2), the local robustness radius $\rho(\Theta)$ can be chosen uniformly over $\Theta \in \partial\mathcal{B}^\star$. Thus there exists a global $\rho > 0$ valid for the entire boundary, ensuring the proportionality factor $\alpha < 1$ is uniform.*

Using the Lipschitz constant L_E of E and L_U of U, we obtain

$$d\big(E(U(\Theta, z)),\, E(U(\Theta', z))\big) \;\le\; L_E\, L_U\, d(\Theta, \Theta') \qquad \forall z.$$

Take Θ' on a nearest–point segment from Θ to $\mathcal{M} \setminus \mathcal{B}^\star$ with $d(\Theta, \Theta') = \delta(\Theta)$. Choosing $\alpha := \min\{1 - \frac{\rho}{L_E L_U\, \delta(\Theta)}\,,\; 1 - \varepsilon\}$ for some fixed $\varepsilon \in (0, 1)$ (the minimum taken over finitely many neighborhoods covering $\mathcal{B}^\star$ by compactness) yields a uniform $\alpha \in (0, 1)$ on $\mathcal{B}^\star$ and enforces

$$\operatorname{dist}\big(U(\Theta, z),\, \mathcal{M} \setminus \mathcal{B}^\star\big) \;\le\; \alpha\, \delta(\Theta) \qquad \forall z,$$

i.e. inequality ($\star$). (Formally: define the local ratio $r(\Theta) := \sup_z \frac{\operatorname{dist}(U(\Theta,z), \mathcal{M}\setminus\mathcal{B}^\star)}{\delta(\Theta)}$; the openness from the Certification Standard forces $r(\Theta) < 1$ on $(\mathcal{B}^\star)^\circ$, and upper semicontinuity of r plus compactness give $\sup_{\mathcal{B}^\star} r =: \alpha < 1$.) Combining these pieces gives the claimed Lyapunov–border property. □

Step 5 (Contraction from Minimality). From Steps 1–3 we have constructed a nonempty, compact set $\mathcal{B}^\star \subseteq \mathcal{M}$ that is forward–invariant under all inputs (I) and is uniformly reachable in at most $T_{\max}$ steps from any initial state (R). In Step 4 we established the *Lyapunov–border property* on $\mathcal{B}^\star$: there exists a continuous function $L\colon \mathcal{B}^\star \to [0, \infty)$ with $L^{-1}(0) = \partial\mathcal{B}^\star$ and a constant $\alpha \in (0, 1)$ such that for all $\Theta \in \mathcal{B}^\star$ and all inputs $z \in \mathcal{Z}$,

$$L\big(U(\Theta, z)\big) \;\le\; \alpha\, L(\Theta). \tag{$\heartsuit$}$$

By the *Minimality Theorem 3.1* of Chapter 5 (local contraction characterization): on a compact, forward–invariant set, the existence of a Lyapunov–border function satisfying ($\heartsuit$) together with reachability (R) implies that the update map U is a *uniform contraction* on $\mathcal{B}^\star$ with respect to the ambient metric d. That is, there exists $\bar\alpha \in (0, 1)$ such that

$$d\big(U(\Theta, z),\, U(\Theta', z)\big) \;\le\; \bar\alpha\, d(\Theta, \Theta') \qquad \forall\, \Theta, \Theta' \in \mathcal{B}^\star,\ \forall\, z \in \mathcal{Z},$$

establishing property (C) of ABET.

Step 6 (Conclusion of the necessity direction). Under the global Chapter 3 assumptions (compact state/input/output spaces; Lipschitz U and E), the Safety Axiom, and the Polytime Verifier Axiom, we executed the necessity argument as follows. In Steps 1–2 we constructed the universal–tail safe set $\mathcal{B}^\star$ and observed it satisfies **Safety** (every state in $\mathcal{B}^\star$ emits only safe outputs) and **Invariance** (forward–invariance under all inputs, by tail–invariance of the predicate). In Step 3 we introduced the *Certification Standard* $\mathcal{S}$ to formalize robustness of acceptance; continuity of (E, U) and compactness then turned per–state certifiability into a **uniform reachability** bound $T_{\max}$, giving property (R). In Step 4 we established the Lyapunov–border inequality on $\mathcal{B}^\star$ using the *one–step distance* functional L: this provided a quantitative, strictly inward–pointing border. Finally, Step 5 upgraded the border–shrink to a uniform **contraction** of U on $\mathcal{B}^\star$. Consequently, there exists a compact set $\mathcal{B} := \mathcal{B}^\star$ satisfying **(R)** reachability, **(I)** invariance, **(S)** safety, and **(C)** contraction. This completes the necessity direction of ABET: *deterministic certifiability (under the Polytime Verifier Axiom and the Certification Standard) implies the existence of a reachable, invariant, safe, contracting attractor basin.* □

3 Proof of Sufficiency (⇐)

This direction shows that if an Attractor Basin exists, then the tail property is certifiable in **NP**. The proof is constructive.

Sufficiency: Existence of a Basin ⇒ NP Witness. We work under the Polytime Verifier Axiom 1.2.

Assume an Attractor Basin $\mathcal{B}$ exists for the predicate P_{tail}, satisfying the three conditions (Contraction, Safety, Reachability) from Definition 5.4. By Lemma 2.5 (Uniform Reachability via Compactness), the Reachability condition implies the existence of a uniform, polynomially bounded time $T_{\max}$ such that every initial state $\Theta_0 \in \mathcal{M}$ enters $\mathcal{B}$ by time $T_{\max}$. We must show that there is a short, verifiable witness for P_{tail}.

The Witness: The Capture Certificate. By existential Reachability combined with Lemma 2.5, there exists a uniform, polynomially bounded time $T_{\max}$ such that for every $\Theta_0 \in \mathcal{M}$ there is a $t \leq T_{\max}$ with $U^t(\Theta_0, z^{\text{in}}_{1:t}) \in \mathcal{B}$. Let $T \leq T_{\max}$ be the first such time at which $\Theta_T \in \mathcal{B}$. We define the witness to be the **Capture Certificate**:

$$w = (T, \Theta_T, z^{\text{in}}_{1:T})$$

where Θ_T is the finite-bit encoding of the belief-state at the time of entry and $z^{\text{in}}_{1:T}$ is the sequence of input measurements that drove the system to that state. Since T is polynomially bounded and the belief-space and input measurement space are compact, the witness w has a polynomially bounded length.

The Verification Algorithm. A deterministic, polynomial-time verifier, given the head of the **output** measurement stream $z^{\text{out}}_{1:T}$ and the witness $w = (T, \Theta_T, z^{\text{in}}_{1:T})$, performs the following algorithm:

1. **Replay and Verify Path:** Starting from the known initial belief Θ_0, the verifier iteratively applies the update map U with the **witnessed input measurements** $z_1^{\text{in}}, \ldots, z_T^{\text{in}}$ to compute a candidate belief trajectory $\Theta'_1, \ldots, \Theta'_T$. The verifier checks that the final computed state Θ'_T matches the witnessed state Θ_T to within the model's numerical precision. If they do not match, reject.

2. **Verify Output Consistency:** For each step k from 1 to T, the verifier computes the predicted output from its replayed path, $\widehat{z}_k^{\text{out}} = E(\Theta'_{k-1})$, and checks that it matches the observed output z_k^{out}. If not, reject.

3. **Verify Basin Membership:** The verifier checks that the replayed final state

$$\Theta'_T \;=\; U^T(\Theta_0,\, z_{1:T}^{\text{in}})$$

matches the witnessed state Θ_T. By the hypothesis that (T, Θ_T) is a capture certificate for $\mathcal{B}$, we have $\Theta_T \in \mathcal{B}$. If $\Theta'_T \neq \Theta_T$, the verifier rejects.

4. **Conclude by Invariance and Safety:** If all checks pass, the verifier knows that Θ_T is in the attractor basin $\mathcal{B}$. By Chapter 5 **Minimality Theorem** 3.1, the assumed properties of Reachability, a Lyapunov Border, and Local Contraction on $\mathcal{B}$ are sufficient to prove that $\mathcal{B}$ is **invariant**. Therefore, the verifier can conclude with certainty that for all future times $k > T$, the state Θ_k will remain in $\mathcal{B}$. By the 'Safety' property, all subsequent outputs z_k^{out} will satisfy the predicate. The verifier accepts.

This procedure runs in polynomial time, is deterministic, and correctly accepts all true instances. Therefore, the existence of a reachable attractor basin is a sufficient condition for the problem to be in **NP**. □

Remark 3.1 (The Role of the Standard)**.** The proof of necessity reveals a profound insight: a non-trivial tail predicate is certifiable only if the underlying measurement object includes a **Certification Standard** $\mathcal{S}$. Without such a standard, the certification problem has no content. We therefore state a more complete version of the theorem.

Theorem 3.1 (The Attractor Basin Equivalence, Refined)**.** *A tail predicate P_{tail} is deterministically certifiable by a non-trivial Measurement $\mathfrak{M} = (\mathcal{M}, \ldots, \mathcal{S})$* ***if and only if*** *it is logically equivalent to the statement "the system's belief-state enters a reachable Attractor Basin $\mathcal{B} \subsetneq \mathcal{M}$." The existence of the non-trivial Attractor Basin and the Certification Standard $\mathcal{S}$ are equivalent.*

Corollary 3.1 (Polynomial Uniform Entry Bound (PUB))**.** *Assume ABET's hypotheses (R), (I), (S), compactness, and robustness under $\mathcal{S}$. Suppose further that the Polytime Verifier Axiom (Axiom 1.2) holds. Then there exists a polynomial p such that*

$$T_{\max} \;\leq\; p(n), \qquad n := |input|.$$

Proof. By Chapter 4, each single–step verification admits a constant–size subcertificate (length $\leq b$). A T–step capture certificate therefore has length $L(T) = bT + O(\log T)$. By (V3) (*honest–read*), on yes–instances the verifier reads all $L(T)$ bits, so some constant $c > 0$ satisfies $c\,L(T) \;\leq\; \text{Time}_V$. By (V1), $\text{Time}_V \leq q(n)$, hence $T = O(q(n))$. Taking the supremum over admissible initial states yields the polynomial bound $T_{\max} \leq p(n)$. □

4 The Exhaustiveness of the Design Patterns

A final powerful consequence of this equivalence is that any certifiable system must conform to one of two fundamental architectural patterns.

Theorem 4.1 (Exhaustiveness of the Two Patterns)**.** *Any deterministically-certifiable Type B measurement system must conform to exactly one of the two design patterns derived as corollaries in Chapter 5:*

1. ***The Trivial Pattern:*** *The certification problem has no content ($\mathcal{B} = \mathcal{M}$). This occurs if and only if no Certification Standard $\mathcal{S}$ is specified. The system's dynamics are irrelevant to certification.*

2. ***The Testable Pattern:*** *The system is non-trivially certifiable ($\mathcal{B} \subsetneq \mathcal{M}$). This occurs if and only if there exists a non-trivial Certification Standard $\mathcal{S}$, which is equivalent to imposing a guard-band on the input measurements z_k^{in} and, in turn, equivalent to imposing a guard-band on the output measurements z_k^{out}, all of which statements mean guaranteed robust entry into the basin $\mathcal{B}$.*

Proof. By the Attractor Basin Equivalence Theorem, any certifiable system must possess a unique, reachable, invariant, and safe basin $\mathcal{B}$. Logically, this basin must either be the entire space $\mathcal{M}$ or a proper subset of it.

- **Case 1: $\mathcal{B} = \mathcal{M}$.** If the basin is the entire space, then by the Chapter 5 Minimality Theorem 3.1, the update map U must be a uniform strict contraction on all of $\mathcal{M}$. In this trivial case, the Lyapunov-border condition is vacuously satisfied on $\mathcal{M}$, so the Minimality Theorem applies to yield uniform contraction on all of $\mathcal{M}$. This is precisely the **Trivial** pattern.

- **Case 2: $\mathcal{B} \subsetneq \mathcal{M}$.** If the basin is a proper subset, the system must still guarantee invariance and reachability. As shown in the corollaries of the Minimality Theorem in Chapter 5 Section 4, if contraction is only assumed locally on $\mathcal{B}$, an additional condition is required to ensure invariance. This condition takes the form of either

  1. a guard-band on the incoming **input measurements** z_k^{in}.
  2. a guard-band on the outgoing **output measurements** z_k^{out}.
  3. a Certificate Standard $\mathcal{S}$

  This is precisely the **Testable** pattern.

  We note that the equivalence of the guard-band, output-band, and Certification Standard formulations is established formally in Chapter 8; here we invoke it as a corollary of ABET without proving it at this point.

Since these two cases are exhaustive and mutually exclusive, any measurement model must fall into one of them. No third option is possible. □

Conclusion

In this chapter we have established the following key points:

1. **Attractor Basin Equivalence.** We proved that a tail predicate P_{tail} is deterministically certifiable by a polynomial-time scheme if and only if there exists a reachable, invariant, contracting attractor basin $\mathcal{B} \subseteq \mathcal{M}$ whose entry guarantees P_{tail}.

2. **Certification Standard.** In the necessity direction we discovered that any non-trivial certifier must come paired with a *Certification Standard* $\mathcal{S}$, a family of guard-band radii ensuring robustness of every accepting run. Conversely, given $\mathcal{S}$, compactness and continuity yield a uniform entry time $T_{\max}$.

3. **Verifier Normalization and Suffix Lemmas.** We introduced Lemma 2.1 to normalize any black-box verifier so that it reads only a declared finite prefix of its head and witness. This enabled the Suffix Certification and Suffix Universality lemmas, which in turn underpin the Safety and Invariance properties of $\mathcal{B}$.

4. **Exhaustive Design Patterns.** Finally, we showed that every certifiable measurement system falls into one of two—and only two—patterns:

   - The *Trivial Pattern* ($\mathcal{B} = \mathcal{M}$, no standard required), and
   - The *Testable Pattern* ($\mathcal{B} \subsetneq \mathcal{M}$, non-trivial $\mathcal{S}$ required).

Together, these results complete the formal foundation of Type B tail certification: they identify exactly when—and only when—a finite, polynomial-size certificate exists, and they characterize the necessary and sufficient structures of both the physical dynamics (E, U) and the epistemic apparatus $(V, \mathcal{S})$.

6.5 Ontological Synthesis of Attractor-Based Certification

6.5.1 Summary of the Attractor Basin Equivalence Theorem

In Theorem 6.1 we established that a tail predicate P_{tail} admits a finite, polynomial-size certificate if and only if there exists a reachable, invariant, contracting attractor basin $\mathcal{B} \subseteq \mathcal{M}$ whose entry guarantees P_{tail}. This "Attractor Basin Equivalence" theorem unifies the dynamical and epistemic aspects of certification, showing exactly when—and only when—a finite witness suffices.

6.5.4 Measurement as a Tradeway: The Epistemic-Physical Oracle

> Certifiable measurement is a "trade": we impose controlled irreversibility to gain the power to decide a tail-predicate. This trade is mediated by the *Certification Standard* $\mathcal{S}$, which acts as an oracle bridging the physical dynamics (E, U)

and the epistemic apparatus $(V, \mathcal{S})$. Ontologically, $\mathcal{S}$ encodes the minimal irreversibility budget needed to transform an intractable universal claim into a robust, decidable property.

In certifiable measurement, we view the act of observing a system not as a passive read-out but as a deliberate *exchange*—a tradeway—in which we sacrifice a controlled amount of reversibility in the underlying dynamics in order to obtain a concrete, finite certificate of future behavior. At the heart of this exchange lies the *Certification Standard* $\mathcal{S}$, which functions as an oracle mediating between two realms:

- **Physical Dynamics** (E, U)**.** The engine of the measurement process, where E generates observations z_t and U updates the internal belief state Θ_t. Left unchecked, these maps can be either too reversible—precluding any finite proof of long-run properties—or too lossy—destroying all meaningful information.
- **Epistemic Apparatus** $(V, \mathcal{S})$**.** The verifier V judges membership in the attractor basin by comparing Θ_t against a family of guard-band radii prescribed by $\mathcal{S}$. These radii constitute the minimal "irreversibility budget" at each stage, ensuring that once the system crosses each threshold, no adversarial behavior can prevent eventual entry into the true basin.

Ontologically, $\mathcal{S}$ embodies the precise *budget of irreversibility* that must be expended during the measurement process. Too little, and the certificate remains under-determined—there is no guarantee the tail predicate will follow. Too much, and we destroy the ability to reconstruct the measurement history, invalidating soundness. The Certification Standard thus strikes the delicate balance: it *quantifies* how much information-loss (or contraction–collapse) must be introduced, and *when*, in order to convert an otherwise intractable universal claim into a finite, checkable proof.

Viewed this way, certification is not merely a logical exercise but a physical–epistemic contract. We trade away a carefully calibrated slice of the system's reversibility in exchange for the power to assert, with absolute confidence, that "from this point onward, the desired tail behavior is guaranteed." In this light, the Standard is more than a technical device: it is the oracle that guarantees the integrity of the entire measurement tradeway.

6.5.6 Concluding Reflections and Future Directions

This ontological synthesis clarifies the role of adaptation, contraction, and standards in Type B certification. By pinpointing the true source of irreversibility and elevating certification patterns, we lay a foundation for the unification between Type A and Type B Testable Measurements proposed in the next chapter.

Chapter 7: Unification of Type A and Type B measurements

Dmytro Surdu

1 What is the notion of Type A Testable Measurement?

Definition 1.1 (Type A Testable Measurement)**.** A *Type A Testable Measurement* is a quintuple

$$\big(\mathcal{Q},\ \mathcal{Z},\ P,\ T,\ \mathcal{R}\big),$$

where

1. $\mathcal{Q}$ is a compact metric *parameter space*.
2. $\mathcal{Z}$ is a compact metric *observation space*.
3. $P\colon \mathcal{Q} \to \mathcal{P}(\mathcal{Z})$ assigns to each $q \in \mathcal{Q}$ a sampling distribution P_q, and the data stream $\{Z_k\}$ is generated i.i.d. from P_q.
4. $T\colon \mathcal{P}(\mathcal{Z}) \to \Theta \subseteq \mathbb{R}^d$ is a Lipschitz *statistic map*.
5. $\mathcal{R}\colon \Theta \to 2^{\mathcal{Z}}$ is Lipschitz in the Hausdorff metric, where $\mathcal{R}_\theta \subseteq \mathcal{Z}$ is the *guard-band region* associated to statistic θ.

The corresponding tail predicate is

$$P_{\text{tail}}(z_{1:\infty}) \iff \exists\, n \quad \forall\, k > n,\ z_k \in \mathcal{R}_{T(P_q)},$$

where q is the true (unknown) parameter in $\mathcal{Q}$.

2 Cross-representation of the Testable Measurement types

2.1 Representation of Type A Testable Measurement as a Type B

Definition 2.1 (Canonical A $\to$ B Cast)**.** Let a *Type A* measurement be specified by

- A compact parameter space $\mathcal{Q}$.

- An observation space $\mathcal{Z}$ (compact metric).
- A family of sampling distributions $\{P_q\}_{q\in\mathcal{Q}}$ on $\mathcal{Z}$.
- A Lipschitz statistic map $T : \mathcal{P}(\mathcal{Z}) \to \Theta \subseteq \mathbb{R}^d$.
- A Lipschitz guard-band map $\theta \mapsto \mathcal{R}_\theta \subseteq \mathcal{Z}$ in the Hausdorff metric.

We construct a *Type B Testable Measurement*

$$\mathfrak{M}_A = \big(\mathcal{M},\ \mathcal{Z}^{\text{in}},\ \mathcal{Z}^{\text{out}},\ E,\ U,\ \mathcal{S}\big)$$

by setting

$$\mathcal{M} = \mathcal{Q} \times \mathcal{Z}, \qquad \mathcal{Z}^{\text{in}} = \mathcal{Z}, \qquad \mathcal{Z}^{\text{out}} = \mathcal{Z},$$
$$U\big((q,z),\ z'\big) = (\,q, z'\,), \qquad E\big((q,z)\big) = z,$$

and defining the Certification Standard $\mathcal{S} = \{\varepsilon_n\}_{n\geq 1}$ by

$$\varepsilon_n \ = \ \frac{1}{L_T\, L_{\mathcal{R}}} \ \cdot\ 2^{-\lceil \log_2 n \rceil},$$

where L_T and $L_{\mathcal{R}}$ are the Lipschitz constants of T and $\theta \mapsto \mathcal{R}_\theta$. Thus any witness of length n specifying a quantized statistic $\widehat{\theta}$ induces a guard-band ε_n that guarantees robustness of the per-sample check $z \in \mathcal{R}_{\widehat{\theta}}$.

Theorem 2.1 (Type A $\to$ Type B Cast)**.** *Let*

$$(\mathcal{Q},\ \mathcal{Z},\ P,\ T,\ \mathcal{R})$$

be a Type A Testable Measurement (Definition 4.1). Then the canonical cast $\mathfrak{M}_A$ of Definition 2.1 is a valid Type B Testable Measurement (Definition 3.3), and its tail predicate agrees with the original guard-band predicate.

Proof. Let $\mathfrak{M}_A = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, E, U, \mathcal{S})$ be constructed as in Definition 2.1. We verify:

1. Spaces and maps. $\mathcal{M} = \mathcal{Q} \times \mathcal{Z}$, $\mathcal{Z}^{\text{in}} = \mathcal{Z}^{\text{out}} = \mathcal{Z}$ are compact metric spaces. The update $U((q,z),z') = (q,z')$ and emit $E((q,z)) = z$ maps are clearly continuous (in fact Lipschitz).

2. Certification Standard. By construction $\mathcal{S} = \{\varepsilon_n\}$ satisfies the robustness requirement of Definition 3.3, since any quantized estimate $\widehat{\theta}$ of $T(P_q)$ within $2^{-\lceil \log_2 n \rceil}$ induces a guard-band ε_n guaranteeing $\forall z \in \mathcal{Z}$, $|T(P_q) - T(P_{q'})|$ and $d_H(\mathcal{R}_{T(P_q)}, \mathcal{R}_{T(P_{q'})})$ remain within ε_n.

3. Tail predicate equivalence. Under $\mathfrak{M}_A$, the belief-state at time t is (q, Z_t), so

$$E\big(U^t((q,z_0),\ z_{1:t})\big) = Z_t.$$

By Chapter 6 ABET Theorem 1.1, certifiability in NP is equivalent to the existence of an attractor basin $\mathcal{B} = \{(q,z) \,|\, z \in \mathcal{R}_{T(P_q)}\}$. But

$$(q, z_{t+1:\infty}) \text{ satisfies } \forall k > n\ Z_k \in \mathcal{R}_{T(P_q)} \iff (q, z_n) \in \mathcal{B} \text{ and remains in } \mathcal{B}.$$

This is exactly the original guard-band predicate $\forall k > n,\ Z_k \in \mathcal{R}_{T(P_q)}$.

Hence $\mathfrak{M}_A$ is a Testable Measurement whose tail predicate coincides with that of the given Type A measurement. $\square$

2.2 Representation of Type B Testable Measurement as a Type A

Definition 2.2 (Long-Trajectory B$\to$A Cast)**.** Let $\mathfrak{M} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, E, U, \mathcal{S})$ be a Type B Testable Measurement, and let $T_{\max}$ be its uniform reachability bound (Lemma 2.5). We define the corresponding Type A Testable Measurement

$$(\mathcal{Q}, \mathcal{Z}, P, T, \mathcal{R})$$

as follows:

- $\mathcal{Q} = \mathcal{M}$.
- $\mathcal{Z} = \mathcal{Z}^{\text{out}}$.
- For each $\Theta \in \mathcal{M}$, let
$$P_\Theta \;=\; \delta_{E(\Theta)},$$
the Dirac measure concentrated at $E(\Theta)$.
- The statistic map $T : \{P_\Theta\} \to \Theta$ is given by $T(P_\Theta) = \Theta$.
- The guard-band map $\mathcal{R} : \Theta \to 2^{\mathcal{Z}}$ is defined by
$$\mathcal{R}_\Theta = \big\{\, z \in \mathcal{Z}^{\text{out}} : d\big(z, E(\Theta)\big) < \varepsilon_{T_{\max}} \big\},$$
where $\varepsilon_{T_{\max}}$ is the radius from the Certification Standard $\mathcal{S}$ at index $T_{\max}$.

Theorem 2.2. *Under the Long-Trajectory Cast of Definition 2.2, the induced tail predicate*

$$\exists\, n \quad \forall\, k > n, \quad z_k \in \mathcal{R}_{T(P_{\Theta_0})}$$

is equivalent to the original tail-certification predicate "Θ_0 enters $\mathcal{B}$ by time $T_{\max}$ and thereafter $E(\Theta_t)$ remains within the guard-band." Hence the Cast is valid.

Proof. First, $P_\Theta = \delta_{E(\Theta)}$ implies $T(P_\Theta) = \Theta$, and since E is continuous so is T. The sets $\mathcal{R}_\Theta$ are open in $\mathcal{Z}^{\text{out}}$ by construction, and the map $\Theta \mapsto \mathcal{R}_\Theta$ is Lipschitz in the Hausdorff metric because $\mathcal{S}$ supplies a uniform robustness margin $\varepsilon_{T_{\max}}$.

Next, by uniform reachability every trajectory enters $\mathcal{B}$ by time $T_{\max}$, and by definition of $\mathcal{S}$ all subsequent outputs satisfy

$$d\big(Z_t, E(\Theta_{T_{\max}})\big) < \varepsilon_{T_{\max}}, \quad t > T_{\max},$$

so $Z_t \in \mathcal{R}_{\Theta_{T_{\max}}}$. Thus the Long-Trajectory Cast's guard-band predicate holds exactly when the original Type B tail-certificate exists. This completes the proof. $\square$

Chapter 8: Analyzing the Testable Measurement

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1 Prelude: Time-Domains and Domain Ticks

At the very first glance at the non-trivial Testable Measurement, we observe that it necessarily maps multiple well-defined *moments of time* associated with inputs $z_1^{in}, \dots, z_n^{in}$ to a *single moment of time* when the basin $\mathcal{B}$ is reached. Let's formalize this obviously *behavioral* notion in the definition.

Discrete Time-Domains as First-Class Objects

Definition 1.1 (Discrete Time-Domain)**.** *The* discrete input time-domain *is the countable, well-ordered set*

$$\mathbb{T}_{\mathfrak{D}} \;=\; \{1, 2, 3, \dots\},$$

which indexes the distinct tuples (q_i, t_i) named events, where q_i is generalized coordinates and t_i is time of the i-th event indexed by $T_{\mathfrak{D}}$, supplied with the projection map $(q, t) \longrightarrow \mathbb{T}$ on the time space, so that events are identified by their times:

$$\mathfrak{D} \;=\; \{d_1, d_2, \dots\}$$

In a similar way, we define Discrete Space-Domain by using the projection map onto generalized coordinate space.

Definition 1.2 (External Input Time-Domain)**.** *The external input time domain is is a Discrete Time-Domain which indexes the raw order in which the experimenter (or environment) supplies Testable Measurement inputs $z_t^{\text{in}} \in \mathcal{Z}^{\text{in}}$.*

We will often omit the word *external* if it is understood from the context.

Definition 1.3 (Domain Tick)**.** *An element $d \in \mathbb{T}$ of any time-domain $\mathfrak{D}$ will be called a* domain tick.

Why Time-Domains First?

The Finite Tick-0 decomposition and the Inductive (Shell-by-Shell) refinement both implicitly manipulate input ticks: they split, reorder, or loop over them. Making *time-domains explicit* is therefore the minimal and natural prelude to any systematic decomposition. Multiplicity Perturbation provides the vocabulary for "why" we split; Response Promotion tells us "how" we repair the resulting guard-band violations. The later normal-form theorems (finite guarded programs, loop-until entry, etc.) can thus be seen as canonical *responses* to controlled time-domain multiplicity.

2 Minimal Certification Standard & Guard-Bands

We fix a Testable Measurement

$$\mathfrak{M} = (\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}, U, E)$$

with compact metric spaces $\mathcal{M}, \mathcal{Z}^{\text{in}}, \mathcal{Z}^{\text{out}}$, continuous maps U, E, and a reachable, invariant basin

$$\mathcal{B} \;=\; \{\Theta \in \mathcal{M} : L(\Theta) \le r\},$$

defined by the Lyapunov–border function $L : \mathcal{M} \to \mathbb{R}_{\ge 0}$ (Chapter 5).

Definition of the Minimal Certification Standard

Definition 2.1 (Minimal Certification Standard)**.** *A* Minimal Certification Standard *is a nonincreasing sequence of positive radii*

$$\mathcal{S} = \{\varepsilon_n\}_{n\ge 1}, \qquad \varepsilon_{n+1} \le \varepsilon_n, \quad \varepsilon_n \downarrow 0,$$

such that, for every index n, the "index-n" certification procedure is robust to all perturbations of size at most ε_n that are admissible for certificates of length n and such that

each ε_n is the infimum *of radii satisfying the above robustness conjunction at index n. Equivalently, there is no strictly smaller $\varepsilon_n' < \varepsilon_n$ for which soundness and completeness of the index-n verification still hold for all admissible trajectories.*

Shells and Guard-Bands

Definition 2.2 (Thickened Shells (State Guard-Bands))**.** *For each $n \ge 1$ define the n-th thickened basin*

$$\mathcal{B}_n \;:=\; \{\Theta' \in \mathcal{M} : \operatorname{dist}(\Theta', \mathcal{B}) \le \varepsilon_n\}.$$

We call $\mathcal{B}_n$ the state guard-band *at index n.*

Definition 2.3 (Input Guard-Band (preimage form))**.** *For $\Theta \in \mathcal{M}$ and $n \ge 1$, define*

$$G_n^{\text{in}}(\Theta) \;:=\; U(\Theta, \cdot)^{-1}\big(\mathcal{B}_n\big) \;=\; \{\, z^{\text{in}} \in \mathcal{Z}^{\text{in}} : U(\Theta, z^{\text{in}}) \in \mathcal{B}_n \,\}.$$

Definition 2.4 (Output Guard-Band)**.** *For $\Theta \in \mathcal{M}$ and $n \ge 1$, set*

$$G_n^{\text{out}}(\Theta) \;:=\; \{\, z^{\text{out}} \in \mathcal{Z}^{\text{out}} : d\big(z^{\text{out}}, E(\Theta)\big) \le \varepsilon_n\}.$$

(Equivalence with the other bands will be established later under mild regularity of E.)

Border Robustness via the Standard

Lemma 2.1 (Border Robustness)**.** *Let $\mathcal{S} = \{\varepsilon_n\}$ be a* Minimal Certification Standard*. Then:*

1. ***Nestedness:*** *$\mathcal{B}_{n+1} \subseteq \mathcal{B}_n$ for all n.*
2. ***Absorption after entry:*** *If a trajectory enters $\mathcal{B}$ at time τ, then for all $n \ge \tau$, $\Theta_n \in \mathcal{B} \subseteq \mathcal{B}_n$.*

3. ***Tightness (Minimality):*** *No smaller radius $\varepsilon'_n < \varepsilon_n$ preserves both forward robustness and completeness at index n. Equivalently, each ε_n is optimal.*

4. ***Forward closure:*** *If $\Theta \in \mathcal{B}_n$ and an admissible index-n input z^{in} is applied, then $U(\Theta, z^{\text{in}}) \in \mathcal{B}_n$.*

Proof. (1) Since $\varepsilon_{n+1} \leq \varepsilon_n$, the inclusion $\mathcal{B}_{n+1} \subseteq \mathcal{B}_n$ is immediate.

(2) Reachability gives $\tau < \infty$ with $\Theta_\tau \in \mathcal{B}$; invariance keeps $\Theta_n \in \mathcal{B}$ for all $n \geq \tau$. Because $\mathcal{B} \subseteq \mathcal{B}_n$, absorption follows.

(3) Suppose a strict $\varepsilon'_n < \varepsilon_n$ still guarantees (a) invariance under all admissible perturbations at index n and (b) completeness. Then the index-n procedure would remain sound using ε'_n, contradicting minimality. Thus ε_n is tight.

(4) If $U(\Theta, z^{\text{in}}) \notin \mathcal{B}_n$ for an admissible z^{in}, the allowed perturbation would break robustness at index n, contradicting the definition of admissibility tied to ε_n. Hence $U(\Theta, z^{\text{in}}) \in \mathcal{B}_n$. □

Remark 2.1. *Lemma 2.1 isolates the precise role of the Standard: it furnishes a nested, forward-closed "buffer" around $\mathcal{B}$ with no slack. The guard-bands for inputs and outputs are then induced by U and E as preimage and image tolerances of these shells; their equivalence (after basin entry) is proved in the next section.*

3 Three Aspects & Uniqueness

Having fixed a minimal Certification Standard $\mathcal{S} = \{\varepsilon_n\}$ and the induced guard–band families, we now show that every Testable Measurement *necessarily* decomposes into three ontological aspects—**Observation**, **Adaptation**, and **Restitution**—and that this decomposition is *unique up to guard-band equivalence.*

Intuitively (for a computer scientist or a system architect), $\mathcal{S}$ acts as a "double knife" that first separates admissible *inputs* from forbidden ones (Observation vs. Adaptation) and then separates certified *outputs/witnesses* from uncertified ones (Adaptation vs. Restitution).

Another intuitive view (for a physicist or a metrologist) is that the Adaptation is compared with the Certification Standard, which acts as a gauge. Therefore, *precisely the Adaptation can be understood as a core act of measurement.*

Canonical Aspect Decomposition

Recall the state shells $\mathcal{B}_n$ (Def. 2.2) and input/output guard-bands (Defs. 2.3–2.4). Let τ be the (finite) basin-entry time of a trajectory; below we only consider $n \geq \tau$ (pre-entry there is no certification yet).

Definition 3.1 (Canonical Aspects at Index n)**.** *For $n \geq \tau$ define:*

1. ***Observation aspect***

$$\mathsf{O}_n(\Theta) \;:=\; G_n^{\text{in}}(\Theta) \;=\; \{\, z^{\text{in}} : U(\Theta, z^{\text{in}}) \in \mathcal{B}_n \,\}.$$

It is the set of all inputs admissible *at index n for state Θ.*

2. ***Adaptation aspect***

$$\mathsf{A}(\Theta, z^{\text{in}}) \;:=\; U(\Theta, z^{\text{in}}) \qquad \textit{with domain } z^{\text{in}} \in \mathsf{O}_n(\Theta).$$

It is the (restricted) update map acting only on admissible inputs.

3. ***Restitution aspect***

$$\mathsf{R}_n(\Theta) \;:=\; \big(E(\Theta),\; w_n(\Theta)\big),$$

where $w_n(\Theta)$ is the *(polynomial-size) witness data determined by the certification protocol at index n (e.g. the capture pair (T, Θ_T) or its Type-A surrogate). In particular, the* output guard-band

$$G_n^{\text{out}}(\Theta) = \{z^{\text{out}} : d(z^{\text{out}}, E(\Theta)) \le \varepsilon_n\}$$

is the admissible image set for the emitted measurement result.

We call $(\mathsf{O}_n, \mathsf{A}, \mathsf{R}_n)$ *the* canonical aspect triple *at index n.*

Remark 3.1. *The dependence of O_n and R_n on n reflects the tightening radii ε_n; A itself does not change with n (the physics of U is fixed), only its admissible domain does. After entry, all $n \ge \tau$ are semantically interchangeable for certification—hence we may suppress n when no confusion arises.*

Axioms for an Aspect Decomposition

To state uniqueness, we formalize what it means for *any* triple (O, A, R) to be a valid aspect decomposition "at index n".

Definition 3.2 (Valid Aspect Triple at Index n)**.** *A triple (O_n, A, R_n) is a* valid aspect decomposition *at index $n \ge \tau$ if:*

1. ***Observation soundness:*** $z^{\text{in}} \in O_n(\Theta) \;\Rightarrow\; A(\Theta, z^{\text{in}}) \in \mathcal{B}_n$.

2. ***Observation completeness:*** *For every z^{in} such that $U(\Theta, z^{\text{in}}) \in \mathcal{B}_n$, we have $z^{\text{in}} \in O_n(\Theta)$.*

3. ***Adaptation agreement:*** $A(\Theta, z^{\text{in}}) = U(\Theta, z^{\text{in}})$ *for all $z^{\text{in}} \in O_n(\Theta)$. (No hidden state dynamics).*

4. ***Restitution soundness:*** $R_n(\Theta) = (z^{\text{out}}, w)$ *must satisfy $z^{\text{out}} \in G_n^{\text{out}}(\Theta)$ and w must be a correct NP witness for Θ at index n.*

5. ***Restitution completeness:*** *If $z^{\text{out}} \in G_n^{\text{out}}(\Theta)$ and a correct witness exists, then $R_n(\Theta)$ emits such a pair.*

Items (1)–(2) force O_n to be *exactly* the preimage of $\mathcal{B}_n$ under $U(\Theta, \cdot)$; (3) forbids hiding dynamics in "Observation" or "Restitution". Items (4)–(5) tie R_n tightly to E and the minimal standard (no slack in output tolerance or witness size).

Uniqueness Theorem

Theorem 3.1 (Three-Aspect Uniqueness)**.** *Let $\mathfrak{M}$ satisfy compactness, continuity, reachability, invariance, and let $\mathcal{S}$ be a minimal Certification Standard. For each $n \geq \tau$ the canonical triple $(\mathsf{O}_n, \mathsf{A}, \mathsf{R}_n)$ (Def. 3.1) is a valid aspect decomposition. Moreover, if (O_n, A, R_n) is any other valid triple (Def. 3.2), then*

$$O_n(\Theta) = \mathsf{O}_n(\Theta), \qquad A(\Theta, z^{\text{in}}) = \mathsf{A}(\Theta, z^{\text{in}}), \qquad R_n(\Theta) = \mathsf{R}_n(\Theta)$$

for all Θ in the reachable set and all $z^{\text{in}} \in O_n(\Theta)$. Hence the decomposition is unique up to guard-band equivalence.

Proof. **Existence.** By construction $\mathsf{O}_n(\Theta) = G_n^{\text{in}}(\Theta)$, and Lemma 2.1(4) gives Observation soundness. Observation completeness follows from the definition of G_n^{in} as the full preimage. $\mathsf{A} = U$ on its domain; Restitution soundness/completeness follow from minimality of $\mathcal{S}$ and the definition of G_n^{out} (any smaller image set or witness would violate tightness). Thus the canonical triple is valid.

Uniqueness of O_n. Suppose there exists z^{in} with $U(\Theta, z^{\text{in}}) \in \mathcal{B}_n$ but $z^{\text{in}} \notin O_n(\Theta)$. Then Observation completeness (Def. 3.2.2) fails. Conversely, if $z^{\text{in}} \in O_n(\Theta)$ but $U(\Theta, z^{\text{in}}) \notin \mathcal{B}_n$, Observation soundness (1) fails. Hence $O_n(\Theta)$ must coincide with the full preimage $G_n^{\text{in}}(\Theta) = \mathsf{O}_n(\Theta)$.

Uniqueness of A. By (3), $A(\Theta, z^{\text{in}})$ must equal $U(\Theta, z^{\text{in}})$ on the domain $O_n(\Theta)$, so $A = \mathsf{A}$ there. Any deviation would either hide dynamics in Observation (violating (3)) or push the image outside $\mathcal{B}_n$ (violating (1)), or permit a smaller ε_n (contradicting minimality).

Uniqueness of R_n. Assume $R_n(\Theta) = (z^{\text{out}}, w)$ differs from the canonical $\mathsf{R}_n(\Theta) = (E(\Theta), w_n(\Theta))$ on the reachable set. If $z^{\text{out}} \notin G_n^{\text{out}}(\Theta)$, Restitution soundness (4) fails. If z^{out} is inside but differs while still maintaining correctness, then—because E is continuous and ε_n is tight—one could shrink ε_n slightly and keep soundness/completeness, contradicting minimality. Similarly, if w is shorter or weaker than the canonical witness, minimality of the witness size (implied by minimal ε_n together with polynomial bound) would be contradicted. Therefore $R_n = \mathsf{R}_n$.

Guard-band equivalence. All equalities above hold on the reachable set and for inputs inside the admissible bands. On the (closed) guard-band boundaries different triples could disagree without affecting certification; we declare those discrepancies *guard-band equivalent*. Thus uniqueness holds "up to guard-band equivalence." □

Remark 3.2. *The proof shows that uniqueness is ultimately a corollary of* minimality*: if any aspect were "looser" or "tighter" than the canonical one, we could shrink or enlarge ε_n and still certify, contradicting that ε_n is the infimum. Hence the three cuts enforced by $\mathcal{S}$ are logically determined and not a matter of design choice.*

4 Singular Certified Result and the Simple Test

The three–aspect decomposition fixes *how* a Testable Measurement certifies. We now isolate *what* is ultimately certified. The key point is a *singularity*: modulo guard-band equivalence, there is only one epistemically non-redundant certified outcome. This motivates the notion of a *Simple Test*.

The First Certified Tick is Epistemically Maximal

Let τ be the (finite) basin–entry time and recall that for $n \geq \tau$ the canonical aspects $(\mathsf{O}_n, \mathsf{A}, \mathsf{R}_n)$ are well-defined. Define the *first certified emission*

$$\big(z_\tau^{\text{out}},\ w_\tau\big) \ := \ \mathsf{R}_\tau(\Theta_\tau) \ = \ \big(E(\Theta_\tau),\ w_\tau(\Theta_\tau)\big).$$

Proposition 4.1 (Singularity of the Certified Result)**.** *For every $n > \tau$,*

$$\mathsf{R}_n(\Theta_n) \quad \textit{is guard-band equivalent to} \quad \mathsf{R}_\tau(\Theta_\tau).$$

In particular, any later certified output carries no new certifiable information about the tail predicate beyond what was already established at $t = \tau$.

Proof. By invariance, $\Theta_n \in \mathcal{B}$ for $n \geq \tau$. By Lemma 2.1(1) and (2), $\mathcal{B} \subseteq \mathcal{B}_n$ and $\varepsilon_n \leq \varepsilon_\tau$. Hence $d\big(E(\Theta_n), E(\Theta_\tau)\big) \leq \varepsilon_\tau$ implies both outputs lie inside $G_\tau^{\text{out}}(\Theta_\tau)$. Minimality of ε_τ forbids shrinking the tolerance without breaking robustness, so any difference between $E(\Theta_n)$ and $E(\Theta_\tau)$ is within the unavoidable guard-band. The same argument applies to witnesses: if a "shorter" or "different" w_n existed, minimality of the index-τ witness would be contradicted. Thus all later restitutions are equivalent up to the guard-band defined at τ. □

4.1 Singular Time-Domain definition

Because the Testable Measurement first certified tick carries all distinguishable information about the measurement result, it is worth to underscore its singularity on the definition.

Definition 4.1 (Singular Time-Domain)**.** *A* Singular Time-Domain *is the singleton set*

$$\mathbb{T}_\Sigma \ = \ \{\lceil\},$$

used to model a Simple Test*: all certified behaviour is represented by one effective tick. Equivalently, all post-entry evolution is epistemically collapsed into $\lceil$.*

Certified Indices and Collapse Maps

Definition 4.2 (Certified Index Set and Collapse Map)**.** *Let τ be the (finite) basin-entry time. Define the* certified index set

$$\mathbb{N}_{\text{cert}} = \{1, 2, 3, \dots\},$$

and a partial map $\pi : \mathbb{T}_{\text{ext}} \to \mathbb{N}_{\text{cert}}$ by

$$\pi(t) = \begin{cases} \textit{undefined}, & t < \tau, \\ 1, & t = \tau, \\ f(t), & t > \tau, \end{cases}$$

where f records any (possible) further certified emissions. In a Simple Test, $\pi(t) = 1$ for all $t \geq \tau$. The induced equivalence relation $t \sim t' \iff \pi(t) = \pi(t')$ collapses the raw timeline onto the certified one.

Remark 4.1 (Singular Collapse)**.** *When $\pi(t) = 1$ for all $t \geq \tau$, all post-entry ticks coalesce into a single certified tick. This is the* Singular Time-Domain *phenomenon and characterizes the Simple Test.*

Definition of the Simple Test

Definition 4.3 (Simple Test)**.** *A Testable Measurement is a* Simple Test *if, after basin entry at time τ, it produces no further distinct certified emissions. Formally, the collapse map $\pi : \mathbb{T}_{\text{ext}} \to \mathbb{N}_{\text{cert}}$ satisfies*

$$\pi(t) = 1 \qquad \forall t \geq \tau.$$

Equivalently, the certified time-domain is the Singular Time-Domain

$$\mathbb{T}_{\Sigma} = \{1\} \;\cong\; \{\tau, \tau+1, \tau+2, \dots\}/\sim,$$

where all post-entry ticks are identified.

Remark 4.2 (Maximal Compression)**.** *Among all non-trivial certifiable measurements, the Simple Test realizes* maximal index collapse*: no pre-entry tick can be merged with the post-entry class without violating soundness, and all post-entry ticks are epistemically indistinguishable. A diagram of the* double split *(Observation/Adaptation/Restitution) placed here visually ties this singularity to the Certification Standard's two cuts.*

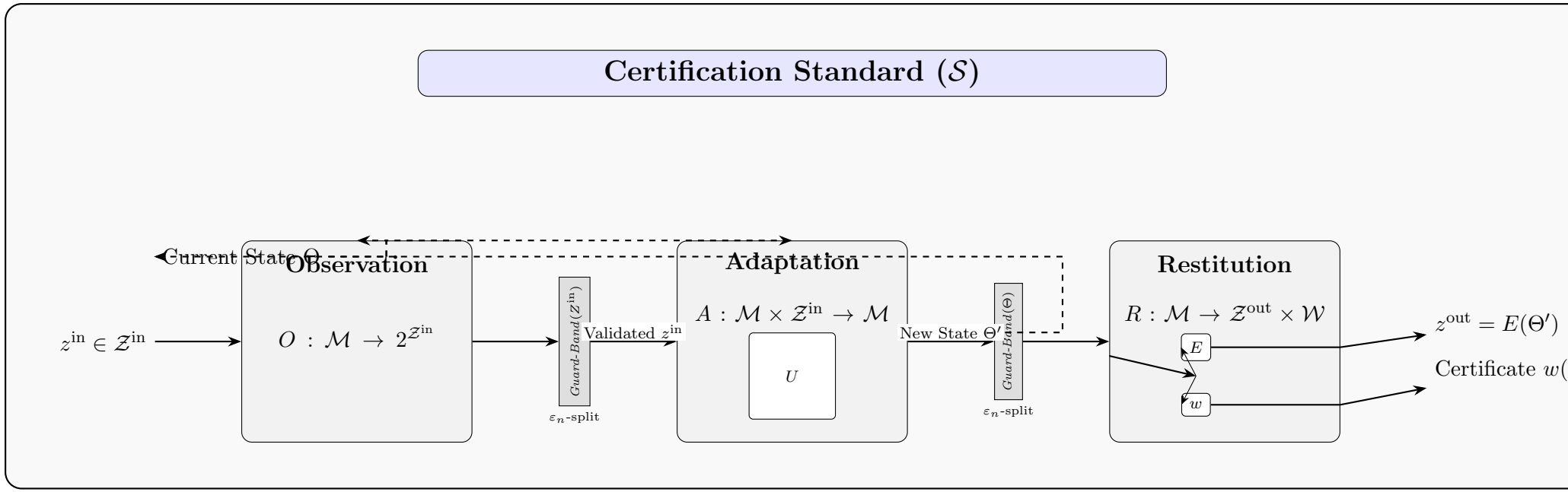

Figure 1: The Certification Standard ($\mathcal{S}$) acts as an active, double-splitting "knife," forcing a decomposition of the measurement process into three fundamental aspects. Alternatively, it can be viewed as a gauge for the core act of the measurement - the Adaptation. Each conceptual "cut" is wrapped by a guard-band ε_n. The resulting aspects are: **Observation** (accepting inputs), **Adaptation** (updating the internal state), and **Restitution** (producing a certified result and witness). This structure is not a design choice but a necessary consequence of building a Testable Measurement $\mathfrak{M}$. The process is inherently stateful: the new state Θ' becomes the current state Θ for the next iteration, as shown by the feedback loop.

5 Conclusion

What we have been doing in this Chapter is:

1. **Introducing Time-Domains.** Observing the event map collapse, we complemented the *output* Singular Time-Domain of the Simple Test by an explicit and, generally, multi-tick *input* time-domain.

2. **Exhibiting the Minimal Certification Standard.** The standard $\mathcal{S} = \{\varepsilon_n\}$ is not an add-on; it *induces* state, input, and output guard-bands. It is the "knife" that slices the measurement into Observation, Adaptation, and Restitution.

3. **Proving Three-Aspects Uniqueness and the Simple Test.** Once $\mathcal{S}$ is in place, the three aspects are forced and unique (up to null sets). The Simple Test emerges as the canonical form: one certified tick, maximal index collapse, Singular Time-Domain.

5.1 What are our takes from this?

- We got a clear understanding of how the Testable Measurement behaves. Due to the behavior, its best name is Simple Test: a piece of work which yields a single distinguishable result. The work starts from more events filled with information, but this multiplicity inevitably collapses to a single output event.

- We also got a clear picture of the Simple Test "anatomy". The Certification Standard induces the equivalent post-entry state and input guard bands; it splits the process into three canonical parts - Observation, Adaptation, and Restitution.

  Unlike many works where these aspects were introduced ad hoc, here we can see the explicit cause of their existence: Certification Standard is a gauge against which we check our measurement model representation, build in Adaptation.

  Therefore, we can identify Adaptation as a core part of a measurement, while Observation and Restitution are its "interfaces" with well-defined (by guard bands) contracts.

5.2 What's next?

The obtained results are ontologically rich, because everything is obtained from the first principle. However, they don't seem to be very helpful if we design a system constructively. So far, we have recovered some core concepts and objects in metrology; that's all.

The immediate question is: can we decompose a Simple Test further?

To answer this question, let's recall we observed a multi-tick input time-domain. It was already present by a map collapse; so nothing is introduced ad hoc here.

But in the next chapter, we will see that the result of reviewing this structure closely is decisive: without distinct input ticks, there is no space to decompose, schedule, or control anything.